UNIT - 1

Introduction to Management

Syllabus at a glance

- Management: concept, nature, functions, science or art
- Schools of thought: Taylor, Fayol, Elton Mayo
- Organizational designs: line, line and staff, functional, matrix, project, committee
- Social responsibilities of management

1. Management - Concept & Meaning

Management is the process of getting things done through others by planning, organising, staffing, directing and controlling the available resources (4 M's: men, money, materials, machines + methods) to achieve pre-determined goals efficiently and effectively.

Definitions (must remember)

- F.W. Taylor : "Management is knowing exactly what you want men to do and seeing that they do it in the best and cheapest way."
- Henri Fayol : "To manage is to forecast and plan, to organise, to command, to coordinate and to control."
- Harold Koontz : "Management is the art of getting things done through and with people in formally organised groups."
- Peter Drucker : Management is a multi-purpose organ that manages business, manages managers and manages workers and work.
- Mary Parker Follett : "Management is the art of getting things done through people."

Meaning in simple words

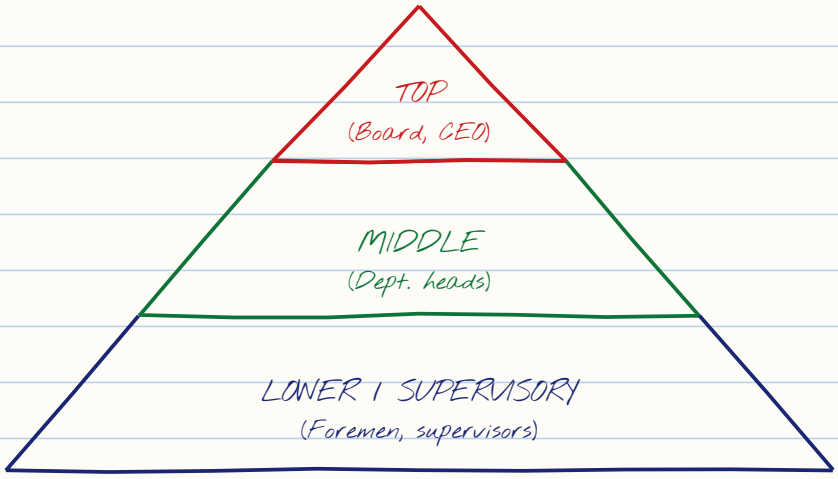
- Management = decision making + resource utilisation + goal achievement
- Efficiency = doing things right (inputs vs outputs)
- Effectiveness = doing the right things (achieving goals)

- A good manager is both efficient and effective

2. Nature (Characteristics) of Management

1. Goal oriented - every management action aims at achieving organisational objectives
2. Universal - needed in business, hospitals, schools, govt., clubs, armed forces
3. Multi-dimensional - management of work, people and operations
4. Continuous process - planning → organising → directing → controlling is a never-ending cycle
5. Group activity - requires team effort and coordination
6. Dynamic function - adapts to changing environment (technology, law, competition)
7. Intangible force - cannot be seen, only felt through results (discipline, profit, morale)
8. Social process - deals with people having feelings, needs and values
9. Authority-responsibility-accountability - manager has the right to command and answers for results
10. Profession or science or art - has body of knowledge, principles and skills

3. Levels of Management & Skills



- Top level : Board of Directors, Chairman, MD, CEO - frame policies, objectives, strategies ; accountable to shareholders

- Middle level : Departmental or functional heads - convert policies into plans and programmes , coordinate between top and lower
- Lower level : Supervisors, foremen, section officers - direct workers , ensure day-to-day output and quality

Robert Katz - three skills of a manager

- Technical skill - Knowledge of methods, tools, processes → most needed at lower level
- Human (interpersonal) skill - leading, motivating, communicating → needed at all levels equally
- Conceptual skill - seeing the organisation as a whole, vision → most needed at top level

4. Functions of Management

The 5 classical functions (Fayol) or 4 modern (Koontz) or POSDCORB (Luther Gulick = Planning, Organising, Staffing, Directing, Co-ordinating, Reporting, Budgeting).

Function	What it means
Planning	Deciding in advance what to do, how, when, by whom . Setting objectives, policies, procedures, rules, budgets, strategies. "Look before you leap"
Organising	Dividing work, grouping activities, assigning duties, delegating authority, establishing relationships → structure
Staffing	Manpower planning, recruitment, selection, training, appraisal, compensation - "right person for the right job"
Directing	Supervision, motivation, leadership, communication - guiding employees to perform
Co-ordinating	Harmonising all activities and efforts; the "essence of management" (Mary P. Follett: reciprocal relating)
Controlling	Setting standards → measuring actual performance → comparing → correcting deviations

Planning-Control link

- Planning is the base, controlling is the check. Planning without control = no use, control without planning = no meaning. They are twin functions ("Siamese twins").

5. Importance of Management

- Achieves group goals - ties individual efforts to common aims
- Increases efficiency - reduces cost, avoids wastage
- Creates dynamic organisation - helps adapt to change
- Achieves personal objectives - motivation, team spirit, good pay
- Develops society - jobs, quality goods, economic growth
- Optimum use of scarce resources

6. Management - Science, Art or Both?

Management as a SCIENCE

- Science = systematic body of Knowledge based on cause-effect relationship, tested by experiments
- Management has organised principles (Fayol's 14, Taylor's principles), developed by observation and experiment (Hawthorne, time study)
- But: principles are not exact like physics, human behaviour is unpredictable, no controlled lab → "inexact or social or behavioural science"

Management as an ART

- Art = personal skill in applying Knowledge to achieve results, needs practice and creativity
- Features of art: theoretical Knowledge, personalised application, based on practice and creativity
- Manager uses communication, leadership, motivation skills in his own style
- Two managers with same Knowledge get different results → skill matters

Management as a PROFESSION

- Profession needs: well-defined body of Knowledge, restricted entry (exam), code of conduct, service motive, representative association
- Management is an emerging profession - lacks compulsory degree or license;

(AIMA, IIMs, ICWA exist)

Conclusion

- Management is both Science and Art - science provides the principles (Knowledge), art provides the application (skill).
- "Science is knowing, art is doing." → A good manager = scientific knowledge + artistic application

Basis	Science	Art
Nature	Systematic knowledge	Personal skill in applying
Based on	Experiment, observation	Practice, creativity
Principles	Universal validity	Personalised
Test	Cause-effect, verifiable	Results achieved
Management	Principles of management	Leadership, communication

7. Management vs Administration

Basis	Administration	Management
Meaning	Determines objectives and broad policies	Executes policies, achieves objectives
Level	Top level	Middle and lower level
Function	Decision-making, thinking	Doing, implementation
Skill	Conceptual	Human and technical
Applies to	Govt., military, big organisations	Business enterprises

8. Schools of Management Thought

Management thought has evolved from ancient times (Chanakya's Arthashastra, Egyptian pyramids) to modern scientific approaches. Main schools:

School	Key persons	Main idea
Scientific Management	F.W. Taylor, Gantt, Gilbreth	Efficiency at shop-floor, "one best way"
Administrative or Classical	Henri Fayol, Urwick, Gulick	Universal principles for the top-level management

School	Key persons	Main idea
Bureaucratic	Max Weber	Rules, hierarchy, impersonality, specialisation
Human Relations or Behavioural	Elton Mayo, Maslow, McGregor	Worker as a human being - social needs and motivation
Quantitative or Management Science	Operations research, Blackett	Maths models, statistics, computers for decisions
Systems approach	Chester Barnard, Bertalanffy	Organisation = open system of inter-related parts
Contingency (Situational)	Lawrence, Lorsch, Fiedler	"It depends" - no one best way; fit to situation

9. Taylor's Scientific Management Theory

Frederick Winslow Taylor (1856-1915) - "Father of Scientific Management". Worked at Midvale and Bethlehem Steel (USA). Book: The Principles of Scientific Management (1911). Aim: replace rule-of-thumb with science to raise productivity.

Four Principles

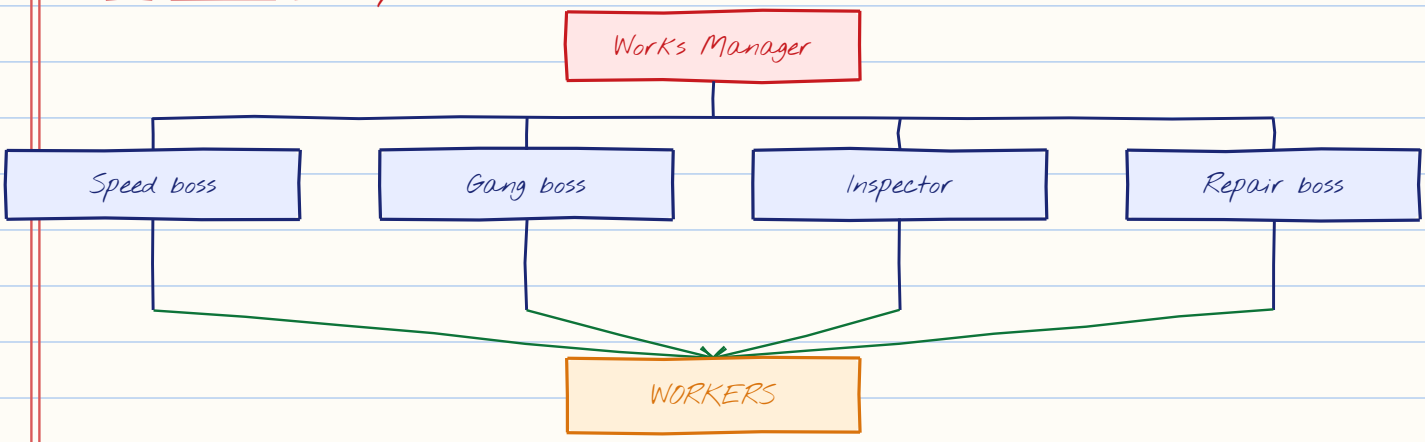
1. Science, not rule of thumb - find "one best way" through time, motion and fatigue study
2. Harmony, not discord - co-operation between workers and management
3. Co-operation, not individualism - mutual trust, no restricted output
4. Maximum output, not restricted output - develop each worker to highest efficiency and prosperity

Techniques of Scientific Management

- Time study : measuring time for each task with a stop-watch → fair day's work
- Motion study (Gilbreth) : eliminating unnecessary movements; "therbligs"
- Fatigue study : rest pauses to reduce tiredness
- Standardisation : of tools, raw material, methods, working conditions
- Differential piece-rate system : higher rate for efficient worker (e.g. Rs.10/unit up to standard, Rs.12/unit above) - penalises low output
- Functional foremanship : 8 specialised foremen (4 in planning dept: route

clerk, instruction card clerk, time and cost clerk, disciplinarian; 4 in production: speed boss, gang boss, inspector, repair boss)

- Scientific selection and training of workers
- Mental revolution : change in attitude of both workers and management - the essence of Taylorism



Taylor's 8 foremen - each worker gets orders from 8 bosses

Merits & Criticism

- Merits: higher productivity, lower cost, better quality, higher wages, scientific selection, standardisation
- Criticism: treats workers as machines (economic man), ignores human and social needs, monotony, unions opposed, violates unity of command (functional foremanship), too much stress on shop floor only

10. Henri Fayol's Administrative Theory

Henri Fayol (1841-1925) - French mining engineer, "Father of Modern (General) Management" . Book: Administration Industrielle et Generale (1916) . Looked at management from the top-level viewpoint.

Six activities of an industrial concern

- Technical (production) - Commercial (buying, selling) - Financial (capital) - Security (protection) - Accounting (records, statistics) - Managerial (planning, organising, commanding, coordinating, controlling)

14 Principles of Management

1. Division of work - specialisation raises efficiency
2. Authority and responsibility - right to give orders goes with accountability

3. Discipline - obedience, respect, agreements
4. Unity of command - one employee, one boss
5. Unity of direction - one head, one plan for a group of activities with same objective
6. Subordination of individual interest to general interest
7. Remuneration - fair to employees and employer
8. Centralisation (and decentralisation) - degree depends on size and nature
9. Scalar chain - line of authority from top to bottom; Gang plank = Fayol's bridge for quick cross-communication
10. Order - "a place for everything and everything in its place" (material order and social order)
11. Equity - Kindness + justice
12. Stability of tenure - avoid high turnover
13. Initiative - freedom to think and execute a plan
14. Esprit de corps - "union is strength", team spirit

Taylor vs Fayol

Basis	Taylor	Fayol
Focus	Shop-floor or operations level	Top management or whole organisation
Approach	Scientist, engineer view	Practitioner, administrator view
Principles	Applied to specific tasks	General and universal
Known as	Father of scientific mgmt	Father of modern mgmt
Unity of command	Violated (8 foremen)	Strictly followed

II. Elton Mayo's Human Relations Approach

George Elton Mayo (1880-1949), Australian psychologist at Harvard. Conducted the Hawthorne Experiments (1924-32) at the Western Electric Company, Chicago, USA - the birth of the Human Relations Movement.

1924-27	1927-29	1928-30	1931-32
Illumination experiments	Relay assembly test room	Mass interview programme	Bank wiring observation room

Phases of Hawthorne studies

1. Illumination experiments - varied light for test and control groups; output rose in both → light was not the only factor
2. Relay assembly test room - 6 girls; rest pauses, hours, pay changed; output kept rising → social/psychological factors (attention, belonging)
3. Mass interview programme - 21,000 employees interviewed; found workers' attitude, informal talk, supervisory style affect output
4. Bank wiring observation room - 14 men; discovered informal groups with own norms ("rate busters", "chiselers") that restrict output

Conclusions

- Social and psychological factors affect productivity more than physical conditions
- Workers are not just "economic men" - they want recognition, belonging, security
- Informal groups exist beside formal organisation and have strong influence
- Supervisory style and participation raise morale → "Hawthorne effect" (behaviour changes when people are observed or given attention)
- Communication and participation important in decision-making

Criticism

- Small sample, poor research design, ignored economic incentives, over-emphasised "happy worker = productive worker", too pro-management

Scientific vs Human relations

- Scientific mgmt: machine model, money motivates, formal organisation
- Human relations: people model, social needs motivate, informal groups matter

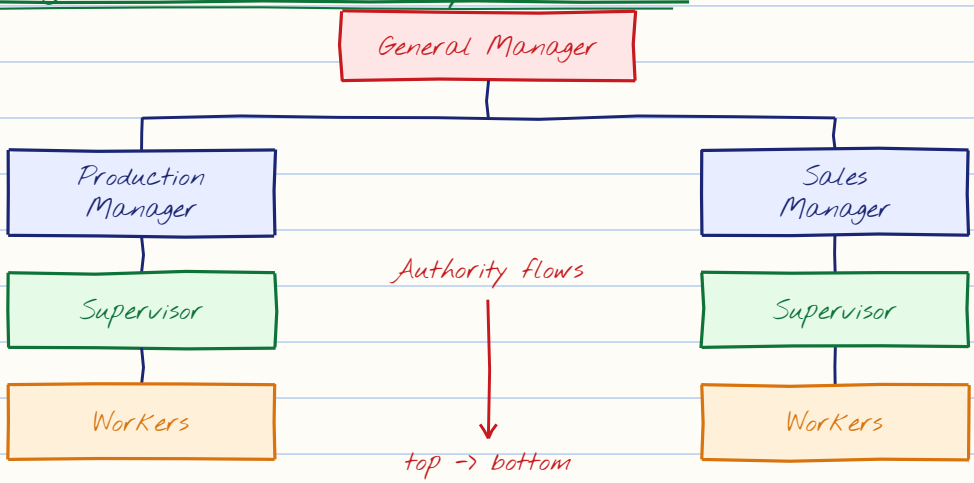
12. Organization & Organizational Designs

Organising = determining activities, grouping them, assigning to people, delegating authority and establishing relationships. An organisation structure is the framework showing who does what and who reports to whom (depicted by an organisation chart).

Principles of organisation

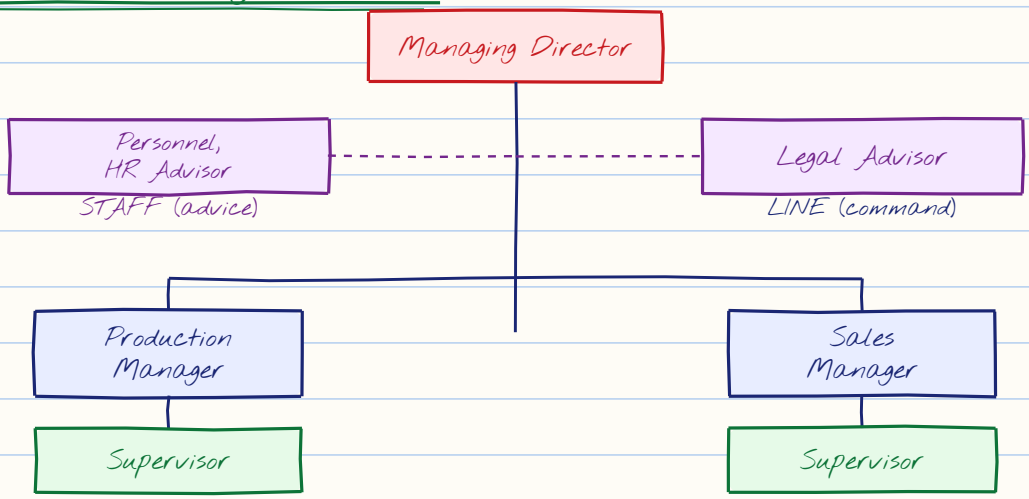
- Unity of objective - Unity of command - Span of control (6-8 at top, 15-20 at lower level)
- Scalar chain - Division of work - Delegation - Balance - Flexibility - Authority = responsibility

(a) Line Organisation - "Military / Scalar"



- Features : direct vertical authority, one boss for one subordinate (unity of command), simple, used in small firms or army
- Advantages : simple, quick decisions, discipline, clear responsibility, economical, unity of command
- Disadvantages : overload on top manager, no specialisation, rigid, autocratic, poor communication, depends on one person
- Suitable for: small and medium units, e.g. a small workshop

(b) Line & Staff Organisation

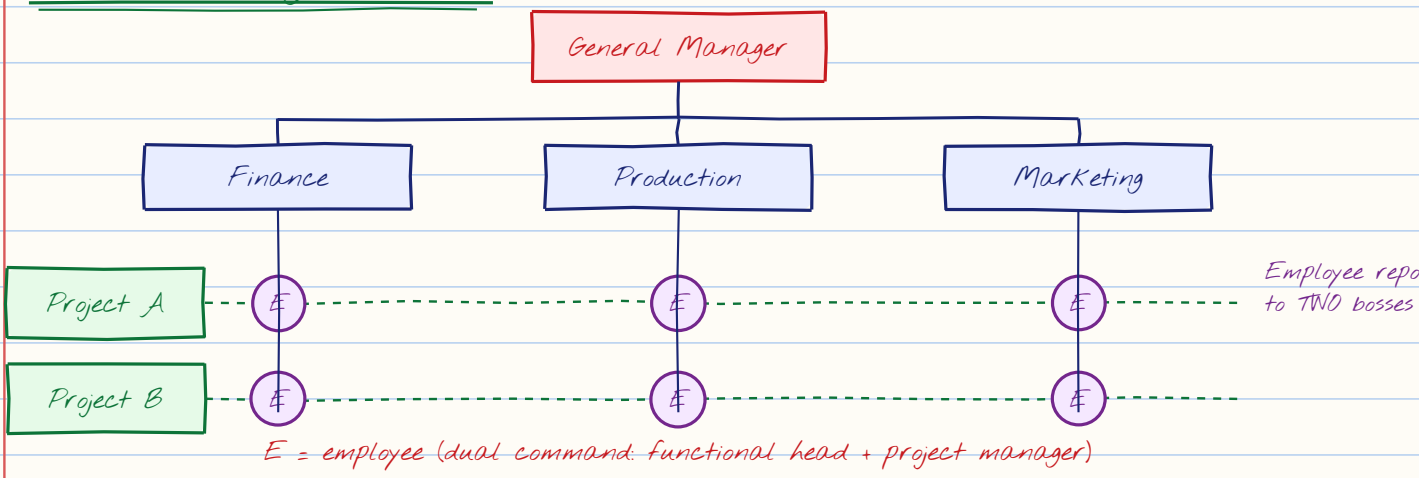


- Line = authority to command and achieve primary objectives, Staff = specialists who advise, help, support the line (no authority to command)
- Types of staff: personal staff (secretary), specialised staff (legal, HR), general staff (planning cell)
- Advantages : expert advice, reduced burden on line, better decisions, balanced growth, flexible
- Disadvantages : line-staff conflict, staff may lack accountability, costly, confusion on authority

(c) Functional Organisation

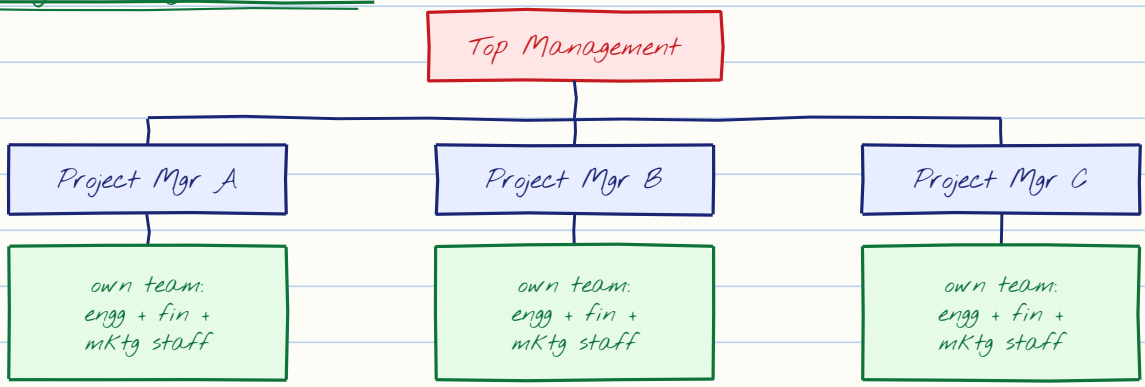
- Based on Taylor's functional foremanship - work is grouped by function (production, marketing, finance, HR) and specialist heads guide workers
- Advantages : specialisation, efficiency, reduced burden, better supervision, expansion easy
- Disadvantages : violates unity of command , confusion and delay, difficult to fix responsibility, inter-department conflict

(d) Matrix Organisation



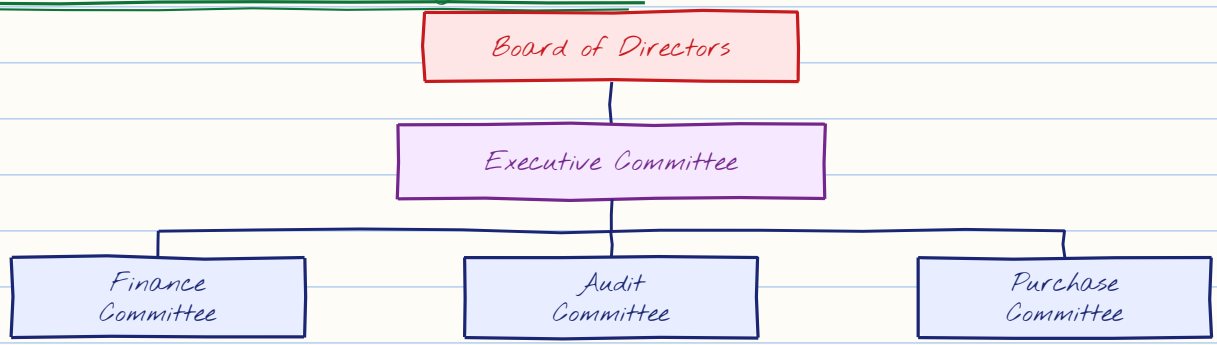
- Combines functional (vertical) + project or product (horizontal) structures; dual command (functional boss + project manager). Used in aerospace, construction, IT, R&D (e.g. NASA, L&T)
- Advantages : efficient use of experts, flexibility, team work, innovation, better project control
- Disadvantages : dual authority → confusion and stress, conflict, costly, needs good communication

(e) Project Organisation



- A temporary structure created for a specific project (dam, bridge, software launch), a project manager has full authority, team members drawn from various departments, dissolved after completion
- Advantages : single focus, speed, clear accountability, skill utilisation
- Disadvantages : duplication of resources, uncertainty after project ends, weak career growth, less specialisation

(f) Committee form of Organisation



- A committee = group of persons to whom a matter is committed for discussion and decision. Types: line or staff, standing (permanent) or ad hoc (temporary), formal or informal, executive, advisory
- Advantages : pooled knowledge and wisdom, group decision, better co-ordination, participation and motivation, checks misuse of power
- Disadvantages : slow, costly, divided responsibility, compromise decisions, "a camel is a horse designed by a committee"

Structure	Key feature	Best suited for
Line	One boss, vertical flow	Small business, army
Line and staff	Line + expert advisers	Medium/large firms
Functional	Specialist bosses	Large, specialised work
Matrix	Dual reporting	Projects + functions together
Project	Temporary team, one manager	Construction, R&D
Committee	Group decision	Policy, co-ordination issues

13. Social Responsibility of Management

CSR = obligation of business to pursue policies, take decisions and follow actions that are desirable in terms of objectives and values of society (Bowen). Business takes resources from society, so it must give back.

Arguments

- For (Keith Davis - Iron Law of Responsibility): long-term self-interest, public image, govt. regulation avoided, better environment, business has resources, "social power creates social responsibility"
- Against (Milton Friedman): business of business is business (profit), dilutes primary goal, costs raise prices, lack of social skills, no accountability to

public

Carroll's pyramid of CSR

- Economic (be profitable) → Legal (obey law) → Ethical (be fair, do right) → Philanthropic (be a good citizen, donate)

Responsibility towards stakeholders

- Shareholders - fair return, true information - Employees - fair wages, safety, training
- Consumers - quality goods, fair price, no adulteration - Government - taxes, law abiding
- Community - jobs, pollution control, education or health programmes - Suppliers and competitors - fair dealing

CSR in India - Companies Act 2013, Section 135

- Applies if company has net worth >= Rs.500 cr or turnover >= Rs.1000 cr or net profit >= Rs.5 cr
- Spend at least 2% of average net profit of preceding 3 years on CSR; form a CSR committee (3+ directors, at least 1 independent)
- Areas (Schedule VII): eradicating hunger and poverty, education, women empowerment, health, environment, rural development, PM Relief Fund, etc.

14. Important Questions (Unit I)

1. Define management. Explain its nature and functions.
2. Is management a science or an art? Discuss.
3. Explain Taylor's scientific management and its techniques.
4. Explain Fayol's 14 principles of management.
5. Discuss the Hawthorne experiments and the contribution of Elton Mayo.
6. Compare Taylor and Fayol.
7. Explain line, line and staff and functional organisation with merits and demerits.
8. Write short notes on: matrix organisation, project organisation, committee organisation.
9. Explain social responsibilities of management.

Last-minute revision

- Taylor = Science, shop floor, 8 foremen, time study, differential piece-rate, mental revolution
- Fayol = 14 principles, 5 functions, top level, unity of command, scalar chain
- Mayo = Hawthorne, informal groups, social needs, Human relations

UNIT - II

Operations Management

Syllabus at a glance

- Plant layout, methods of production, work study, SQC, material management (EOQ, ABC)
- Marketing management: concept, functions, marketing mix, channels, advertisement and sales promotion, PLC strategies

A. OPERATIONS MANAGEMENT

1. Meaning & scope

Operations (Production) Management = planning, organising, directing and controlling the activities that convert inputs (men, material, machines, money) into outputs (goods and services) at the right quality, quantity, time and cost.

- Inputs → Transformation process → Outputs (with feedback and control)
- Scope: product design, plant location, plant layout, production planning and control, quality control, material handling, maintenance, work study
- Objectives: right quality, right quantity, right time, right cost (the 4 R's)

2. Plant Layout

Plant layout = arrangement of machines, equipment, materials, workers and service areas within the factory building so that production is smooth, economical and safe.

Objectives

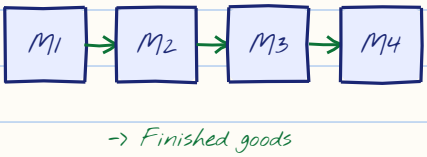
- Smooth flow of material and minimum handling - Better use of floor space - Reduce production cycle time
- Safety and comfort of workers - Flexibility for future changes - Reduce WIP inventory and cost

Principles of a good layout (Moore)

1. Integration - combine men, material, machine in one system
2. Minimum distance - shortest path for material movement
3. Flow principle - arrange in the order of operations, avoid back-tracking
4. Cubic space - use vertical space too (not only floor)
5. Satisfaction and safety - for workers
6. Flexibility - easy rearrangement

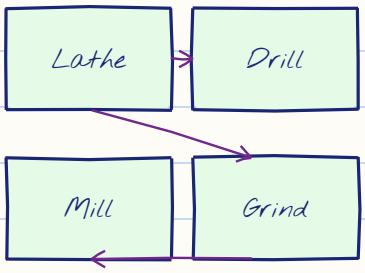
Types of layout

PRODUCT (line) layout



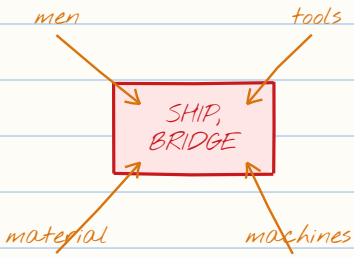
one product, fixed sequence

PROCESS (functional) layout



similar machines grouped, many products

FIXED-POSITION layout



product stays, resources move

Type	Merits	Demerits
Product (line)	Low handling cost, less WIP, simple control, fast output, less skilled labour	Costly, inflexible, one breakdown stops all line, monotony
Process (functional)	Flexible, general-purpose machines, easy to handle breakdown, skilled workers' growth	More handling, higher WIP, difficult control, long cycle time
Fixed position	No movement of heavy product, flexible, good for large products	Costly movement of men and tools, low machine use, space needed
Cellular (Group technology)	Combines product and process benefits, shorter set-up, high flexibility	Needs part families, more planning

Which layout where?

- Product layout → mass production (cars, bottling, sugar) - Process layout → job or batch (machine shops, hospitals)
- Fixed position → ships, aircraft, bridges, dams - Group layout → families of similar parts (cells)

3. Methods (Types) of Production

Basis	Job production	Batch production	Mass (flow) production
Meaning	One or few units as per customer order	Limited quantity produced in lots or batches	Large standardised volume, continuous
Product	Non-standard, custom-made	Semi-standard	Highly standard
Machines	General purpose	General purpose + some special	Special purpose, automatic
Layout	Process or fixed position	Process or group	Product layout
Labour	Highly skilled	Semi-skilled	Low skilled
Cost per unit	Highest	Medium	Lowest
Flexibility	Maximum	Moderate	Minimum
Examples	Ships, bridges, tailor-made suits, dams	Bakery, medicines, books, shoes	Cars, cement, sugar, refrigerators, bulbs

- Continuous (process) production : non-stop flow, e.g. oil refining, chemicals, fertilizers - also a type of mass production
- Intermittent production = job + batch ; Continuous = mass + process

4. Work Study

Work study = systematic examination of methods of carrying out activities so as to improve effective use of resources and set standards of performance (ILO).

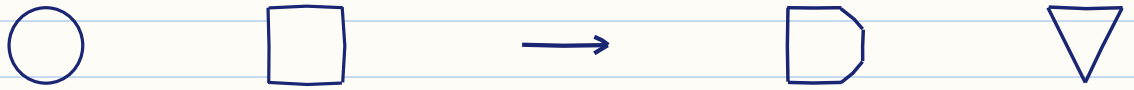
Two parts: Method study + Work measurement. Aim: higher productivity with no extra capital.

(A) Method Study - "Find the best way"

Steps - SREDDIM

1. Select the job to be studied 2. Record all facts (charts, diagrams) 3. Examine critically (what? why? where? when? who? how?)
4. Develop the best method 5. Define the new method 6. Install it 7. Maintain it

- Recording tools: Outline process chart, Flow process chart, Flow diagram, Man-machine chart, Multiple-activity chart, SIMO chart
- Critical examination - ECRS: Eliminate, Combine, Rearrange, Simplify



Operation Inspection Transport Delay Storage

(B) Work Measurement (Time study)

Work measurement = determining time required by a qualified worker to do a task at a defined level of performance. Techniques: stop-watch time study, work sampling, synthetic data (PMTS, MTM), standard data.

- Steps of time study: select job → record method → break into elements → time with stop-watch (several cycles) → rate the worker → add allowances → compute standard time

Formulae

- Normal time = Observed time × (Rating % ÷ 100)
- Standard time = Normal time + Allowances = Normal time × (1 + Allowance %)
- Allowances: personal needs, fatigue, delay, contingency (total about 10–20%)

Example

- Observed time = 2.0 min, performance rating = 110%, allowances = 15%
- Normal time = 2.0 × 1.10 = 2.20 min
- Standard time = 2.20 × 1.15 = 2.53 min per unit → in an 8-hour shift

(480 min) expected output = 480 ÷ 2.53 = ~190 units

5. Statistical Quality Control (SQC)

Quality = fitness for use or conformance to specification. SQC = use of statistical techniques (sampling, control charts) to monitor and control the quality of a process or product. Pioneer: Walter A. Shewhart (1924, Bell Labs).

Causes of variation

- Chance causes (random, natural) - small, unavoidable, process is "in control" e.g. slight temperature changes
- Assignable causes (special) - due to tool wear, wrong setting, bad material, operator error - can be identified and removed → process "out of control"

Two major tools

- Acceptance sampling : inspect a sample from a lot and accept or reject the lot. Risks: producer's risk (alpha, good lot rejected), consumer's risk (beta, bad lot accepted). Terms: AQL, LTPD, OC curve, sampling plan (single, double, multiple)
- Process control (control charts) : plot sample measure against time with 3-sigma limits (UCL, CL, LCL)

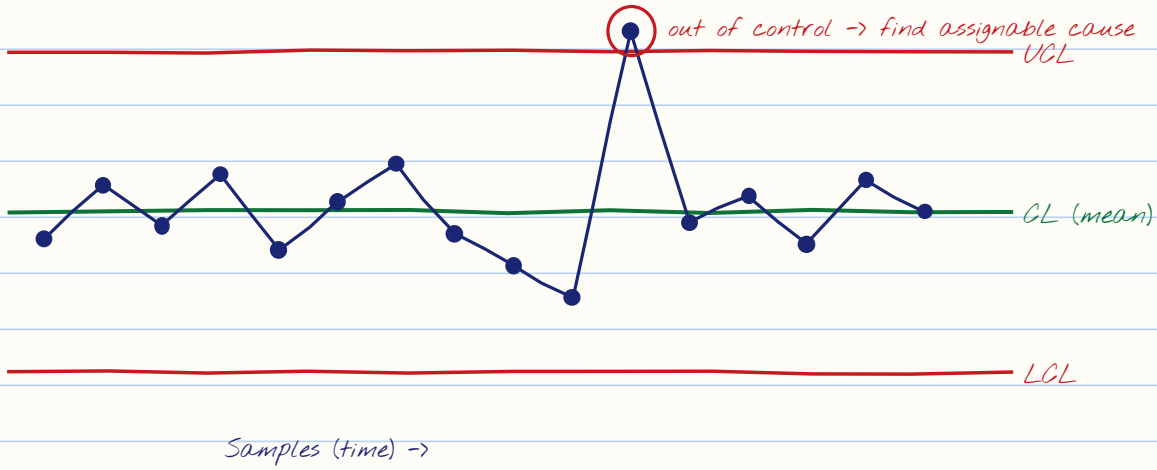


Chart	Type	Used for
X-bar and R chart	Variables	Mean and range of measurable data (length, weight)
p chart	Attributes	Fraction defective (variable sample size)
np chart	Attributes	Number of defectives (fixed sample size)

Chart	Type	Used for
c chart	Attributes	Number of defects per unit (cloth flaws, scratches)

Control limit formulae

- X-bar chart : UCL = X-bar + A2 × R-bar , LCL = X-bar − A2 × R-bar (n = 5 → A2 = 0.577)
- R chart : UCL = D4 × R-bar , LCL = D3 × R-bar (n = 5 → D4 = 2.114, D3 = 0)
- p chart : p-bar ± 3 × sqrt(p-bar (1 − p-bar) ÷ n) ; c chart : c-bar ± 3 × sqrt(c-bar)

Example

- X-bar = 50 mm, R-bar = 4 mm, n = 5 : UCL = 50 + 0.577×4 = 52.31 , LCL = 50 − 2.31 = 47.69 ; R chart UCL = 2.114×4 = 8.46 , LCL = 0
- c chart, c-bar = 9 defects: UCL = 9 + 3×3 = 18 , LCL = 9 − 9 = 0

- Benefits of SQC: early detection of faults, less rework and scrap, lower inspection cost, better customer satisfaction, basis for TQM

6. Material Management

Materials management = planning, organising and controlling the flow of materials from identification of need to the point of delivery to production (purchasing, stores, inventory, handling, disposal). Materials form 50–60% of the cost of a product - so a small saving matters a lot!

Objectives

- Right material, right quantity, right quality, right place, right time, right price (the "6 R's")
- Minimum inventory and storage cost - Continuous supply to production - Good vendor relations - Standardisation and value analysis - Reduce wastage, obsolescence

Functions

- Material planning and budgeting - Purchasing (vendor selection, tenders, negotiation) - Receiving and inspection - Storing and stores accounting -

Inventory control - Material handling - Disposal of scrap and surplus

7. Inventory

Inventory = stock of materials held for future use or sale: raw materials, components, WIP, finished goods, spares, consumables.

Functions (why hold stock?)

- Ensures continuous production - Protects against uncertainty in supply or demand (buffer)
- Gets quantity discounts and saves ordering cost - Hedge against price rise or inflation - Smooths seasonal demand (anticipation stock) - Better customer service

Types of inventory

- Raw materials - Work-in-progress (WIP) - Finished goods - MRO (maintenance, repair, operating supplies) - Spares and tools
- By role: Cycle stock , Safety (buffer) stock , Anticipation stock , Pipeline (transit) stock , Decoupling stock

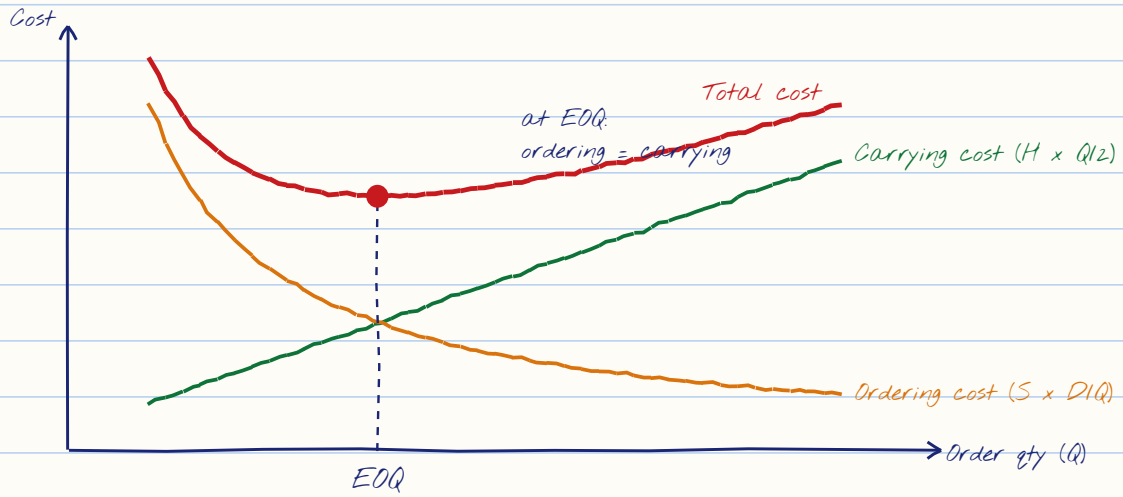
Inventory costs

- Ordering (procurement) cost - paperwork, transport, inspection (per order)
- Carrying (holding) cost - interest, storage, insurance, obsolescence, pilferage (15-30% of value per year)
- Shortage (stock-out) cost - lost sales, idle labour, goodwill - Purchase cost - price × quantity

8. Inventory Techniques

(a) Economic Order Quantity (EOQ) - Wilson's formula

The order size that gives the lowest total of ordering + carrying cost . At EOQ: ordering cost = carrying cost.



Formula

- EOQ = sqrt(2 x D x S ÷ H) where D = annual demand, S = ordering cost per order, H = carrying cost per unit per year (= i x unit price)
- Orders per year N = D ÷ EOQ ; Cycle time = 12 months (or 360 days) ÷ N ; Total cost = sqrt(2 x D x S x H)
- Assumptions: constant demand, constant lead time, no stock-outs, no quantity discount, instant replenishment

Example

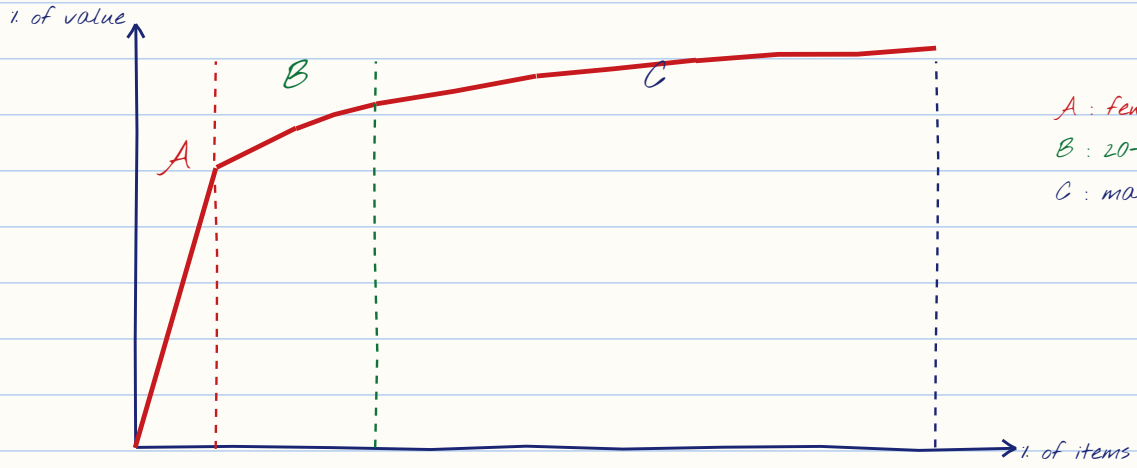
- D = 3600 units/year, S = Rs.200 per order, H = Rs.4 per unit/year
- EOQ = sqrt(2 x 3600 x 200 ÷ 4) = sqrt(360000) = 600 units
- Orders per year = 3600 ÷ 600 = 6 ; interval = 2 months
- Ordering cost = 6 x 200 = Rs.1200 ; Carrying cost = (600 ÷ 2) x 4 = Rs.1200 ; Total = Rs.2400 (equal costs - verified!)

Stock levels

- Reorder level (ROL) = maximum usage x maximum lead time (or = lead-time demand + safety stock)
- Minimum level = ROL - (normal usage x normal lead time) ; Maximum level = ROL + ROQ - (min usage x min lead time)
- Danger level = normal usage x emergency lead time ; Safety stock = buffer to cover uncertainty

(b) ABC Analysis (Always Better Control)

Based on Pareto's 80-20 rule - classify items by annual consumption value (annual usage x unit price) and control each class differently.



Class	% of items	% of value	Control policy
A	10-20%	70-80%	Tight control, accurate records, frequent review, small safety stock, careful vendor selection, EOQ
B	20-30%	15-20%	Moderate control, periodic review, reasonable records
C	50-70%	5-10%	Simple control, bulk ordering, large safety stock, minimum paperwork

- Steps: list items → annual usage value → arrange in descending order → compute cumulative % → divide into A, B, C
- Other classifications: VED (vital, essential, desirable), FSN (fast, slow, non-moving), HML (high, medium, low price), SDE (scarcity), JIT or Kanban (just-in-time)

B. MARKETING MANAGEMENT

9. Concept & Meaning

Marketing = human activity directed at satisfying needs and wants through exchange processes (Kotler). AMA : "Marketing is the process of planning and executing conception, pricing, promotion and distribution of ideas, goods, services to create exchanges that satisfy individual and organisational goals."

Era or concept	Focus	Slogan
Production	Availability, low cost	"Make more"
Product	Quality, features	"Better product sells"

Era or concept	Focus	Slogan
Selling	Aggressive promotion	"Sell what you make"
Marketing	Customer needs and profit	"Make what sells"
Societal	Customer + society welfare	"Customer, company, community"

Nature of marketing

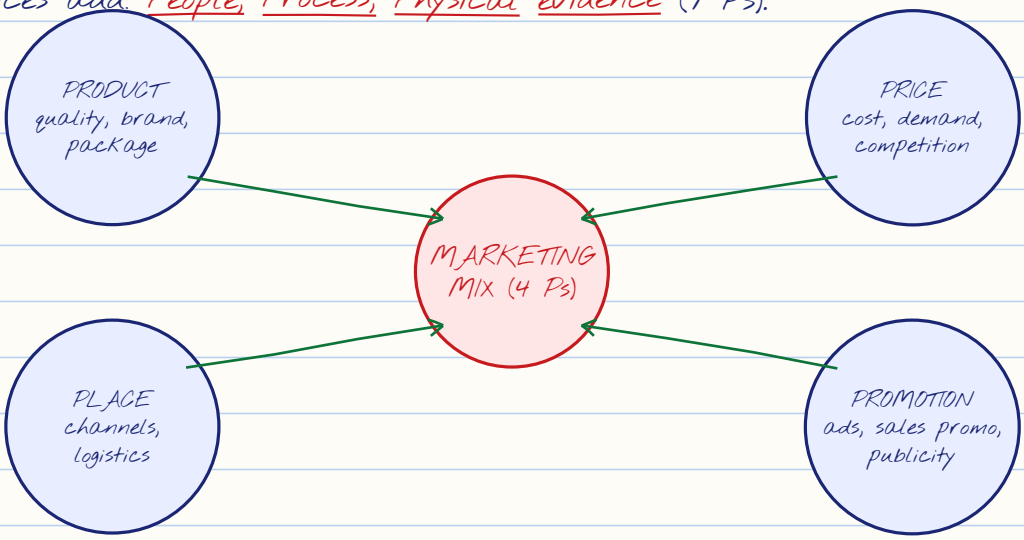
- Needs-oriented - Exchange process - Customer-centred - Both a science and an art - Dynamic and continuous - Creates 4 utilities: form, place, time, possession

Functions of marketing

- Exchange functions : buying, selling - Physical functions : transport, storage, packaging
- Facilitating functions : financing, risk-taking, grading and standardisation, market information
- Others : market research, product planning, pricing, promotion, distribution, after-sales service
- Difference: Marketing = need-based, whole process (before and after sale), Selling = product-based, ends with the sale

10. Marketing Mix

Neil Borden coined, E. Jerome McCarthy gave the 4 Ps. Marketing mix = controllable variables combined to get desired response from the target market. For services add: People, Process, Physical evidence (7 Ps).



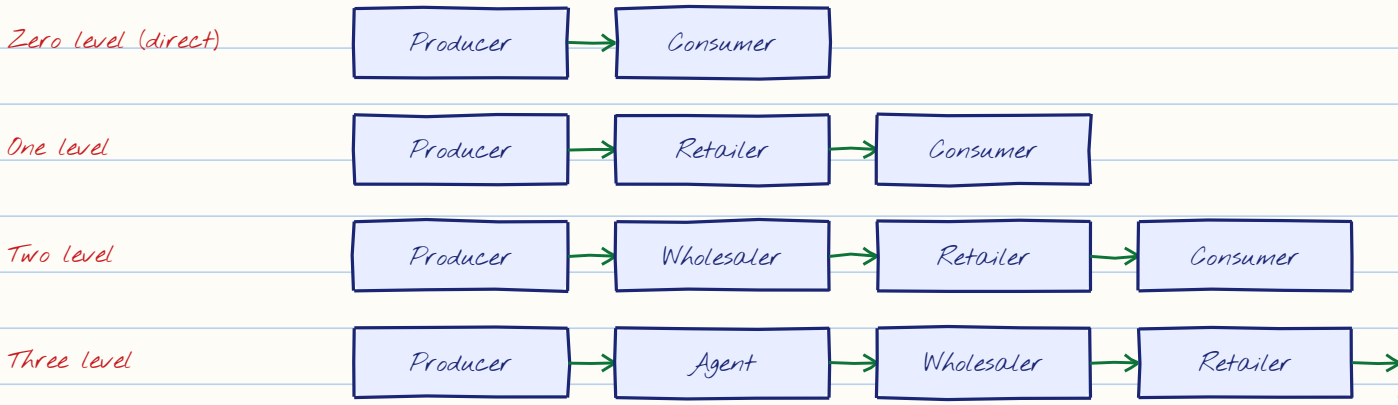
- PRODUCT : features, quality, design, brand, packaging, labelling, warranty,

product range, product life cycle

- PRICE : list price, discounts, credit terms. Methods: cost-plus, break-even, competition-based, value-based, penetration (low) and skimming (high) pricing
- PLACE : distribution channels, coverage, inventory, transport, warehouses, outlets
- PROMOTION : advertising, personal selling, sales promotion, publicity or PR, direct marketing
- Marketing strategies need consistency between the Ps (a luxury brand = premium product + high price + exclusive outlets + prestige ads)

II. Channels of Distribution

A channel = path or chain of intermediaries through which goods move from producer to consumer. Intermediaries: wholesalers, retailers, agents, brokers, distributors, C&F agents .



Factors for choosing a channel

- Product (perishable → short, technical → direct) - Market (size, geography, buyers' habits) - Company (finance, control wanted)
- Middlemen (availability, cost) - Competition - Cost or margin of channel - Legal and environmental factors

- Modern channels: e-commerce, franchising, direct selling, vertical marketing systems (VMS), omni-channel retail

12. Advertisement & Sales Promotion

Advertising

Advertising = paid, non-personal presentation of ideas, goods or services by an identified sponsor through mass media. Goals: inform, persuade, remind; build brand image.

- Media: Print (newspapers, magazines), Broadcast (TV, radio), Outdoor (hoardings, transit), Digital (social media, search, influencer, e-mail), cinema
- Merits: wide reach, creates demand, brand building, low cost per contact - Demerits: costly, no instant feedback, impersonal, may mislead (ASCI controls ads)
- Model - AIDA : Attention → Interest → Desire → Action

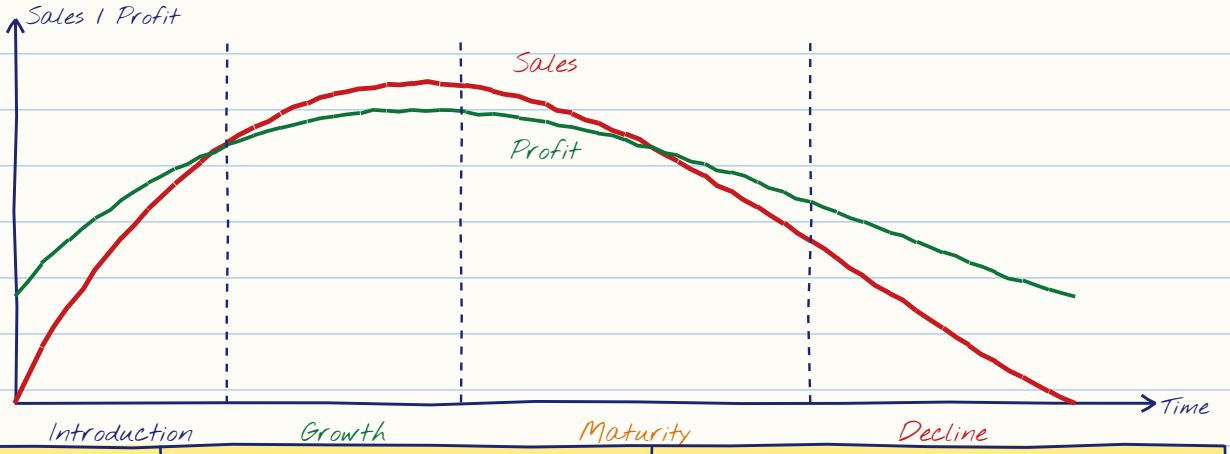
Sales promotion

- Consumer promotion : free samples, coupons, price-off, buy-1-get-1, contests, gifts, cashback, loyalty points, exchange offers
- Trade promotion : trade discounts, dealer contests, display allowances, trade shows
- Sales-force promotion : bonus, commission, sales contests, incentive trips
- Advertising = long-term, pull strategy ; Sales promotion = short-term, immediate sales ; Personal selling = face-to-face

Design an advertisement (practical steps)

- 1. Set objective and target audience 2. Fix budget 3. Create message (USP, headline, slogan) 4. Choose media 5. Design (visual, colour, layout, call-to-action) 6. Test and launch 7. Measure effect (recall, sales)
- Example ("Fresh Sip" juice): Headline "100% fruit, 0% worry" ; Visual: fruit splash ; Offer: buy 2 get 1 free this month ; CTA: scan QR to order ; Media: Instagram + college hoardings

13. Marketing Strategies by Product Life Cycle (PLC)



Stage	Features	Marketing strategy
Introduction	Low sales, high cost, losses, few competitors	Heavy promotion to create awareness, skimming or penetration price, limited distribution, trial offers
Growth	Rapid sales and profit rise, competitors enter	Improve quality, add features, build brand preference, widen distribution, reduce price a little
Maturity	Peak sales, slowing growth, intense competition	Product modification, market modification, new uses, discounts, defend share, loyalty programmes
Decline	Sales and profit fall, obsolescence	Harvest or divest, cut cost and promotion, drop weak products, serve loyal niche, or reinvent

14. Important Questions (Unit II)

1. Define plant layout. Explain its objectives, principles and types.
2. Compare job, batch and mass production.
3. What is work study? Explain method study and work measurement.
4. Explain SQC and the control charts (X-bar, R, p, c) with an example.
5. Define inventory. Explain its functions and types. Derive EOQ and solve a problem.
6. Explain ABC analysis with a neat graph.
7. Explain marketing mix. Discuss marketing strategies at different stages of

PLC.

8. Discuss channels of distribution and sales promotion methods.

Last-minute revision

- EOQ = sqrt(2DS ÷ H) ; ordering cost = carrying cost at EOQ ; ABC = 80-20 rule ; SREDDIM ; UCL = X-bar + A2R-bar ; 4Ps ; AIDA ; Intro-Growth-Maturity-Decline

UNIT - III

Human Resources Management (HRM)

Syllabus at a glance

- HRM meaning, nature, managerial and operative functions - Job analysis - HRP
- Recruitment and selection - Training and development - Performance appraisal
- Placement, induction - Wage and salary administration

1. HRM - Definition & Meaning

Human Resource Management (HRM) is the part of management that deals with procuring, developing, motivating, utilising and maintaining a competent workforce to achieve organisational goals.

- Edwin Flippo : "HRM is planning, organising, directing and controlling of the procurement, development, compensation, integration, maintenance and separation of human resources to the end that individual, organisational and societal objectives are accomplished."
- Dale Yoder : HRM is the management of manpower - a strategic and coherent approach to the management of an organisation's most valued assets: the people.
- Evolution: Personnel management (welfare, clerical) → Personnel and IR → HRM (strategic, people as assets) → HRD

Nature of HRM

- Pervasive (every organisation, every manager) - People-oriented - Continuous - Action-oriented (solves problems, not just records)
- Integrating (individual + organisation) - Dynamic - Strategic partner of top management - Multi-disciplinary (psychology, sociology, law, economics)

Objectives & importance

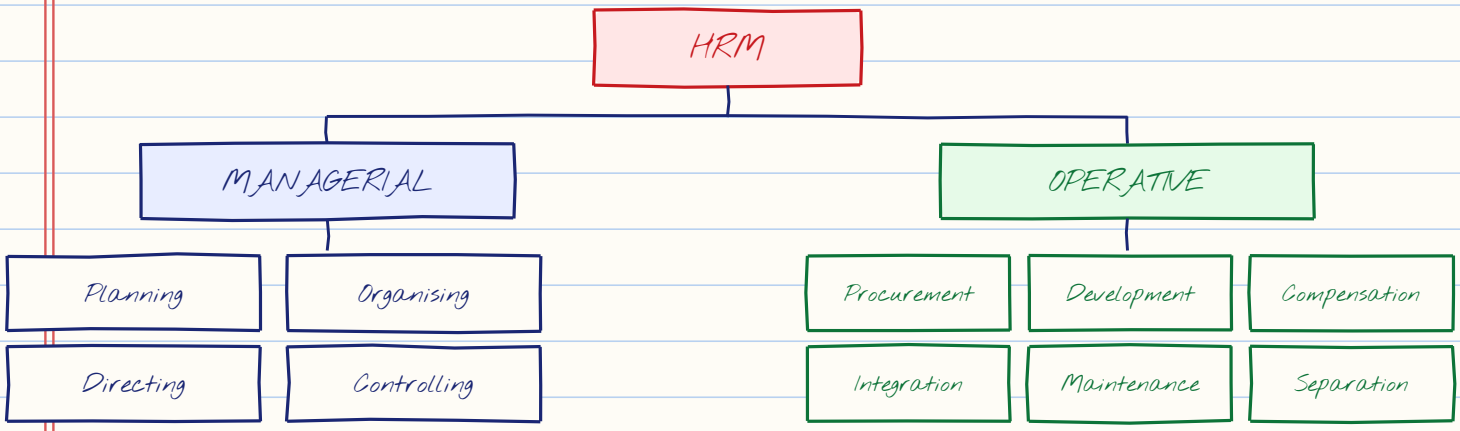
- Societal objectives (ethical, social responsibility) - Organisational (right people, right job, productivity) - Functional (department's contribution) - Personal

(growth, satisfaction)

- Importance: attracts and retains talent, improves productivity, reduces turnover and disputes, builds culture, ensures legal compliance

Basis	Personnel management	HRM
View of people	Cost or tool	Asset or resource
Approach	Reactive, routine, clerical	Proactive, strategic, developmental
Focus	Rules, procedures, welfare	Commitment, empowerment, performance
Role	Staff function	Line + staff, top-level involvement

2. Functions of HRM - Managerial & Operative



Managerial functions

- Planning - forecast manpower, set HR policies, programmes, budgets - Organising - HR department structure, duties, delegation, relationships
- Directing - motivation, leadership, communication, supervision - Controlling - compare performance with standards, audits, corrective action

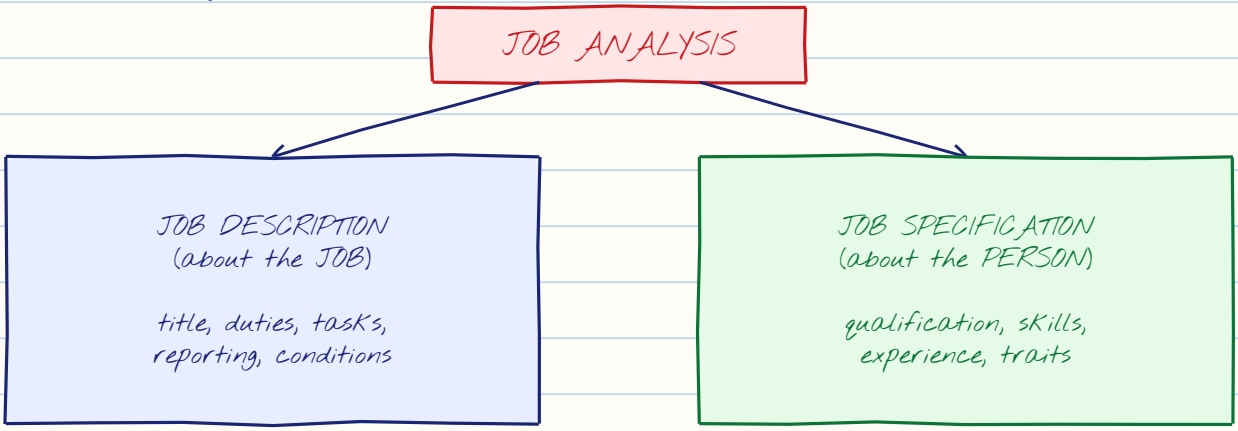
Operative functions

Function	Activities
Procurement	Job analysis, HR planning, recruitment, selection, placement, induction
Development	Training, executive development, career planning, performance appraisal

Function	Activities
Compensation	Job evaluation, wage and salary fixing, incentives, bonus, fringe benefits
Integration	Employee relations, grievance handling, collective bargaining, participation, motivation
Maintenance	Health, safety, welfare, social security, morale, working conditions
Separation	Resignation, retirement, retrenchment, dismissal, VRS, exit interview

3. Job Analysis

Job analysis = systematic process of collecting and recording information about the duties, responsibilities, skills, working conditions and requirements of a job. (What does the job involve? Who can do it?)



Process

- Identify purpose → collect background info (org charts) → select jobs → collect job data → prepare job description and specification → review and update

Methods of collecting job data

- Observation (direct, for repetitive jobs) - Interview (with job holder and supervisor) - Questionnaire (structured form) - Work diary or log - Technical conference (experts) - Critical incident technique - Combination (best)

Sample Job Description (Production Supervisor)

- Title : Production Supervisor ; Dept : Assembly ; Reports to : Plant Manager
- Duties : plan daily output, assign work, ensure quality and safety, maintain records
- Specification : B.Tech or Diploma, 3 yrs experience, leadership, communication, Knowledge of MS Office

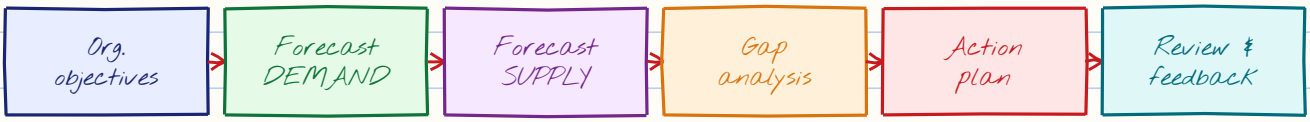
- Uses: recruitment and selection, training needs, performance appraisal, job evaluation (pay), HR planning, safety, career planning

4. Human Resource Planning (HRP)

HRP = process of forecasting an organisation's future demand for and supply of the right type of people in the right number at the right time (Geisler).

Objectives

- Ensure optimum use of HR - Meet future manpower needs - Avoid surplus or shortage - Control labour cost - Support expansion, technology change and succession planning



Surplus -> retrench, VRS, redeploy | Shortage -> recruit, train, overtime, outsource

Process of HRP

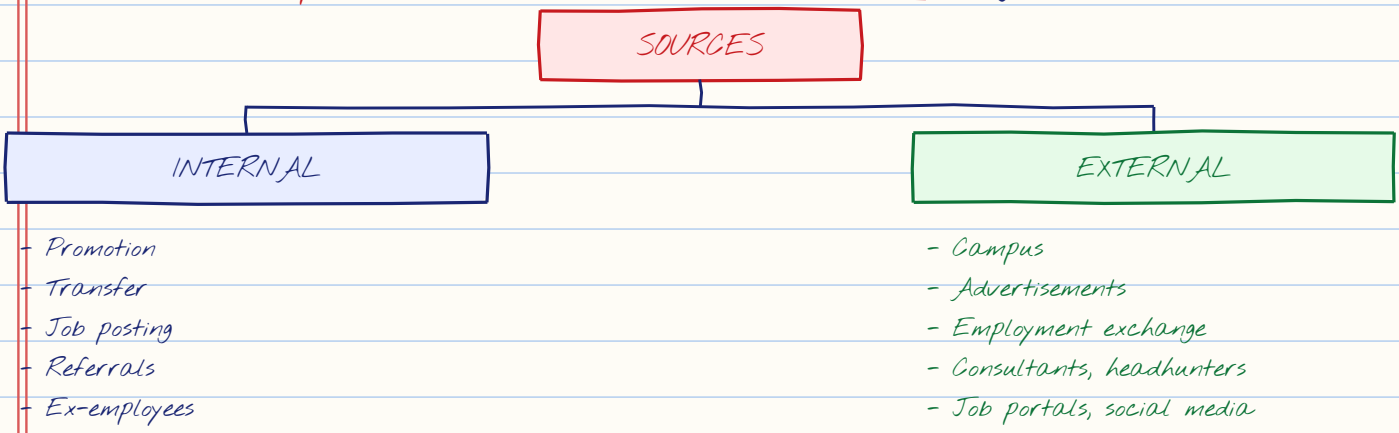
1. Analyse organisational objectives and plans (growth, new products, technology)
2. Forecast demand : managerial judgement, Delphi technique, ratio-trend analysis, regression, work-study (workload) analysis, zero-base
3. Forecast supply : internal - skill inventory, replacement chart, Markov analysis, succession plans ; external - labour market, migration, education output
4. Gap analysis : Demand - Supply = surplus (+) or shortage (-)
5. Action plans : recruitment, training, redeployment, VRS, outsourcing, job redesign

6. Implement and evaluate - audit, update

- Barriers: inaccurate forecasts, resistance from managers or unions, lack of information system, uncertainty, costs

5. Employee Recruitment

Recruitment = process of searching for prospective candidates and stimulating them to apply for jobs (Flippo). Selection = choosing the right ones from the pool. (Recruitment is positive - attracts; selection is negative - rejects.)



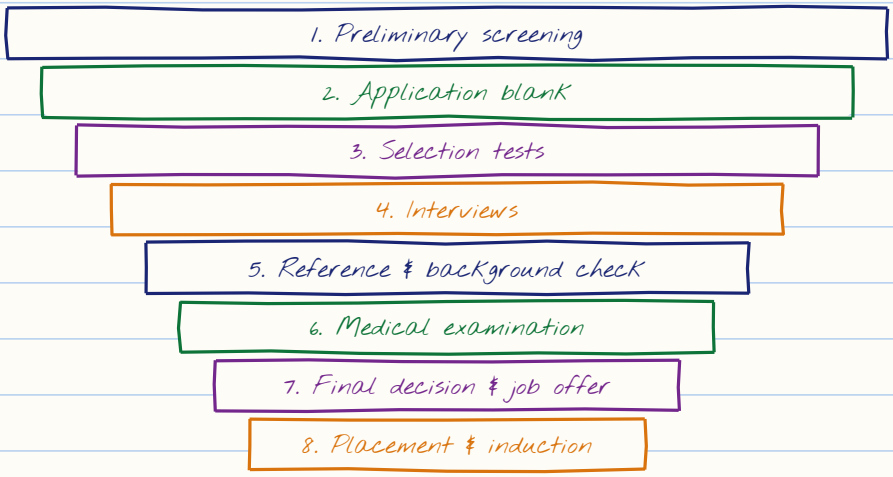
Source	Merits	Demerits
Internal	Cheap, quick, motivates staff, known performance, less training	Limited choice, no fresh ideas, office politics, inbreeding
External	Fresh talent, wider choice, new skills and ideas	Costly, time-taking, risk of poor fit, demotivates insiders

External sources in detail

- Campus recruitment - fresh graduates from colleges (IITs, IIMs) - Advertisements - newspapers, trade journals - Employment exchanges (govt.)
- Placement agencies or consultants and headhunters for senior jobs - Walk-ins and write-ins - Job portals or e-recruitment (NaukRi, LinkedIn) - Referrals - Labour contractors - Unsolicited applications

6. Employee Selection - Process

Selection = "the process of differentiating between applicants to identify those with greater likelihood of success in a job". Multiple hurdles are used.



- Preliminary screening : quick check of eligibility to remove unsuitable candidates
- Application blank : bio-data, qualifications, experience, references
- Tests : aptitude, intelligence (IQ), personality, interest, trade or skill, projective, achievement ; must be valid and reliable
- Interview : face-to-face evaluation - structured, unstructured, panel, stress, behavioural, group
- Reference and background check - Medical fitness - Final selection by line manager → offer and appointment letter

Recruitment vs Selection

Basis	Recruitment	Selection
Nature	Positive - attracts many candidates	Negative - rejects unsuitable ones
Aim	Create a large pool of applicants	Choose the best fit from the pool
Order	First step	Follows recruitment
Output	Applications received	Appointment of selected candidates

Selection tests - types

Test	What it measures	Example
Intelligence	Reasoning, memory, numerical and verbal ability	IQ test
Aptitude	Potential to learn a skill	Mechanical, clerical, programming aptitude
Personality	Traits, emotional stability, attitude	MBTI, Big-5, 16PF

Test	What it measures	Example
Interest	Preference for types of work	Strong Interest Inventory
Trade or proficiency	Present skill level	Typing speed, welding, coding test
Projective	Hidden feelings, imagination	Rorschach inkblot, TAT

Types of interview

Type	Description
Structured (patterned)	Same fixed questions to all - fair, easy to compare
Unstructured	Free-flowing, open conversation - flexible but biased
Panel or board	A group of interviewers - joint and balanced decision
Stress	Pressure created deliberately to test composure (sales, defence)
Behavioural	"Tell me about a time when ..." - past behaviour predicts future
Group or telephonic or video	Many candidates together ; remote screening

Placement & Induction

- Placement = putting the selected person on the right job (matching specification with skills and interests) ; Induction (Orientation) = introducing the new employee to the organisation, people, policies, culture
- Induction contents: company history and vision, products, structure, rules, benefits, safety, tour, buddy system. Benefits: lower anxiety, quick adjustment, reduced turnover

7. Employee Training & Development

Basis	Training	Development
Meaning	Improves skills for a present job	Overall growth for future roles
Focus	Technical, short-term	Conceptual, long-term, career
Level	Mostly operatives	Managers or executives
Initiative	Employer	Self-motivated

Systematic training process

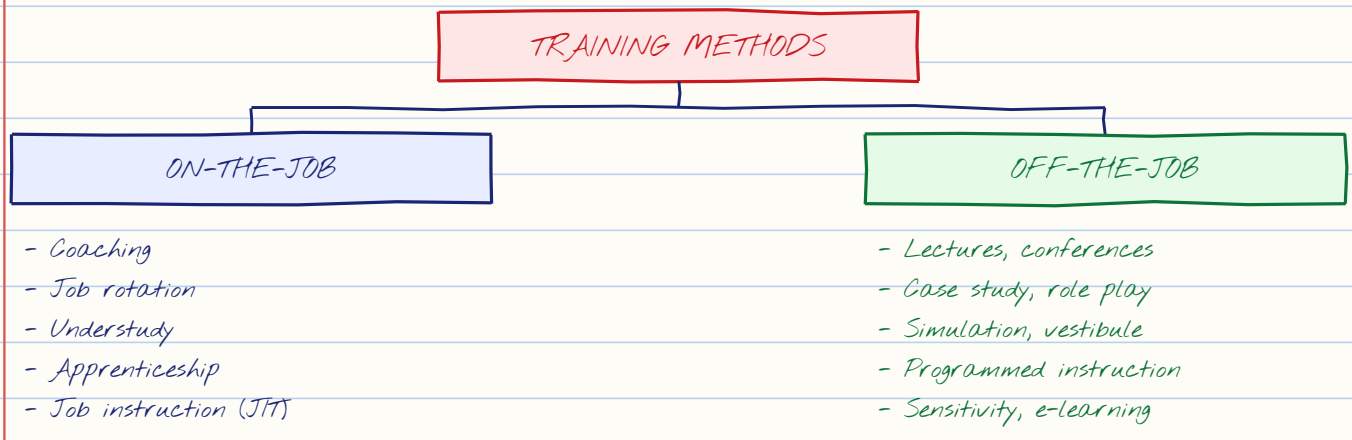
1. Training need analysis (TNA) 2. Set objectives 3. Design programme and methods 4. Implement 5. Evaluate (Kirkpatrick: Reaction → Learning → Behaviour → Results)

Analyse the need for training (3 levels)

- Organisational analysis : goals, resources, skills gap, new technology, high rejection or accidents
- Task (operational) analysis : what knowledge and skills each job demands (from job description)
- Person analysis : actual vs required performance - from appraisal, tests, complaints, absenteeism, supervisor feedback
- Signs of need: low productivity, wastage, more accidents, customer complaints, new machines, promotions, change in methods

Evaluation of training - Kirkpatrick's 4 levels

Level	What is measured	How
1. Reaction	Did trainees like it?	Feedback forms
2. Learning	Knowledge or skill gained	Pre-test vs post-test
3. Behaviour	Applied on the job?	Supervisor observation
4. Results	Productivity, quality, cost, profit	Business metrics, ROI



- Coaching (supervisor guides) - Job rotation (moves across jobs) - Understudy (assistant to a senior) - Apprenticeship (earn + learn) - Vestibule (replica of shop-floor)
- Case study - Role play - In-basket - Business games - Sensitivity training

(T-group) - Programmed or e-learning - Lecture or conference

- Development: mentoring, job enrichment, action learning, MDP, sabbaticals, 360-degree feedback, succession planning

8. Performance Appraisal

Performance appraisal (PA) = systematic evaluation of an employee's job performance and potential against set standards, for development and decisions (promotion, pay, transfer, training).

Objectives & process

- Pay rise and promotion decisions - Identify training needs - Feedback and motivation - Improve communication - Validate selection
- Process: set standards → communicate → measure actual performance → compare → discuss (feedback) → take action (reward or train or correct)

Methods of appraisal

Class	Method	Idea
Traditional	Ranking, paired comparison	Rank best to worst ; compare each with every other
Traditional	Grading, forced distribution	A, B, C grades ; fixed % in each category (bell curve)
Traditional	Graphic rating scale	Rate traits (quality, initiative) on 1-5 scale
Traditional	Checklist, essay, critical incidents	Yes or no statements ; narrative ; record good and bad events
Modern	MBO (Drucker)	Jointly set targets, review results
Modern	BARS	Rating scale anchored with behaviour examples
Modern	360-degree	Feedback from boss, peers, subordinates, customers, self
Modern	Assessment centre	Exercises, role plays, tests by assessors
Modern	HR accounting, 720 or KPI-BSC	Value of employee ; balanced scorecard

Evaluating the appraisal system - errors to avoid

- Halo effect (one good trait colours all) - Horn effect - Central tendency (everyone average) - Leniency or strictness - Recency bias - Personal bias, stereotyping
- Good appraisal = valid, reliable, job-related, objective, participative, regular, linked to development

Appraisal interview (feedback) - good practice

- Be specific, use facts and examples not personality - Start with strengths - Discuss problems and agree an action plan - Set next targets (SMART) - Keep records, follow up

Sample appraisal form (graphic rating scale, 1 = poor, 5 = excellent)

- Job Knowledge 1 2 3 4 5 - Quality of work 1 2 3 4 5 - Quantity of work 1 2 3 4 5
- Initiative 1 2 3 4 5 - Team work 1 2 3 4 5 - Attendance and punctuality 1 2 3 4 5
- Overall rating = total ÷ 6 ; Remarks of reviewer : Training needs : Promotion potential :

9. Wage & Salary Administration

Wage = remuneration to workers (hourly or daily or piece) ; Salary = fixed monthly pay to clerical or managerial staff. Wage and salary administration = scientific fixing of pay structure that is fair, competitive and affordable.

Types of wages

- Minimum wage (basic needs; Minimum Wages Act 1948 → now Code on Wages 2019) - Fair wage - Living wage - Nominal vs real wage (purchasing power)

Factors affecting pay

- Job evaluation (relative worth) - Labour market rates - Ability to pay - Cost of living - Productivity - Govt. laws and unions - Employee skills or seniority

Job evaluation methods

- Non-analytical : Ranking , Job classification (grading) ; Analytical : Point rating (points to factors - skill, effort, responsibility, conditions), Factor comparison

Point-rating job evaluation (example)

- Factors and maximum points : Skill 100 ; Effort 60 ; Responsibility 80 ; Working conditions 40 (total 280)
- Job "Machine Operator" scores Skill 60 + Effort 40 + Responsibility 40 + Conditions 30 = 170 points ; "Supervisor" scores 90 + 40 + 70 + 20 = 220 points
- Pay grades are fixed by point ranges (e.g. 150–180 = Grade II, 200–230 = Grade III)

Methods of wage payment

System	Formula or working	
Time rate	Wage = hours worked × rate per hour. Simple, but no incentive	
Piece rate	Wage = units produced × rate per piece. Encourages output, quality risk	
Taylor's differential	80% (low rate) below standard ; 120% (high rate) at or above standard	
Halsey premium	Earnings = (T × R) + 50% × (time saved × R) (T = actual time)	
Rowan	Earnings = (T × R) + (time saved ÷ time allowed) × T × R	
Bonus, profit sharing	Share of surplus ; Payment of Bonus Act 1965 (8.33% – 20%)	

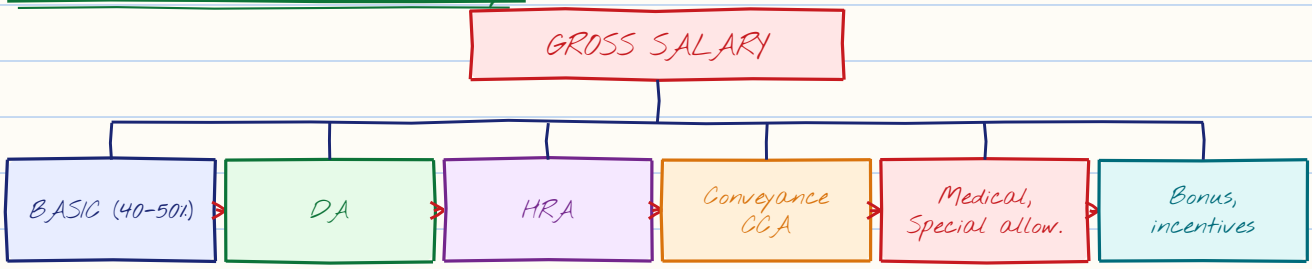
Example

- Standard time 10 hrs, actual 8 hrs, rate Rs.20 per hour (time saved = 2 hrs)
- Time rate = 8 × 20 = Rs.160 ; Halsey = 160 + 0.5 × 2 × 20 = Rs.180 ; Rowan = 160 + (2 ÷ 10) × 8 × 20 = Rs.192
- Taylor: standard 10 units/day, piece rate Rs.10 → 12 units: 12 × 12 = Rs.144 ; 9 units: 9 × 8 = Rs.72

Incentives & fringe benefits

- Financial incentives : bonus, profit sharing, commission, stock options (ESOP), gain sharing
- Non-financial : recognition, job security, promotion, flexible hours, training chances
- Fringe (indirect) benefits : PF, gratuity, pension, medical and insurance, housing, canteen, leave travel, education, creche
- Incentive plan conditions : simple, fair, attainable, quick payment, linked to output and quality, union agreement

Basic structure of salary



minus deductions: PF, ESI, professional tax, TDS = NET (take-home) pay

Component (annual CTC Rs.6,00,000)	Amount (Rs.)	Note
Basic salary (40%)	2,40,000	Base for PF, gratuity, HRA
HRA (50% of basic)	1,20,000	House rent allowance
Special allowance	1,99,700	Conveyance, medical, etc. (balancing figure)
Employer PF (12% of basic)	28,800	Retirement benefit
Gratuity (about 4.8% of basic)	11,500	After 5 years of service
Total CTC	6,00,000	Take-home is lower after deductions

10. Important Questions (Unit III)

1. Define HRM. Explain its nature, scope and functions.
2. Explain managerial and operative functions of HRM.
3. What is job analysis? Distinguish job description and job specification.

4. Explain HR planning and its process.
5. Discuss the sources of recruitment. Explain the selection process.
6. Explain training methods. How do you analyse the need for training?
7. Discuss the methods of performance appraisal. Evaluate their errors.
8. Explain wage payment systems (Halsey, Rowan, Taylor) with example.
9. Short notes: induction, placement, job evaluation, salary structure.

Last-minute revision

- HRM functions : Procurement – Development – Compensation – Integration – Maintenance – Separation
- Selection = screening → blank → tests → interview → check → medical → offer ; Training = on-the-job or off-the-job ; PA = traditional or modern (360, MBO, BARS)

UNIT - IV

Strategic & Project Management

Syllabus at a glance

- Vision, mission, goals - Corporate planning process - Environmental scanning - Strategy formulation and implementation - SWOT
- Project management - Network analysis - PERT and CPM - Critical path - Probability of completion - Project cost analysis and crashing

1. Strategic Management - Meaning

A strategy is a long-term, comprehensive plan of action to achieve organisational objectives by matching the organisation's resources with its environment. (Greek: strategos = "art of the general".)

- Alfred Chandler : "Strategy is the determination of the basic long-term goals of an enterprise and the adoption of courses of action and allocation of resources necessary to carry out these goals."
- Glueck : a unified, comprehensive and integrated plan relating the strategic advantages of the firm to environmental challenges
- Strategic management = continuous process of formulating, implementing and evaluating strategies to achieve long-term success

Features of strategy

- Long-term and future oriented - Top-management decision - Matches resources with environment - Gives competitive advantage - Involves major resource commitment - Flexible and reviewed

Levels of strategy

Level	Who or what	Example
Corporate	Board or CEO - which businesses to be in (growth, diversify, retrench)	Tata group entering EVs
Business (SBU)	Business head - how to compete in one business	Low-cost vs premium positioning

Level	Who or what	Example
Functional	Dept. heads - marketing, finance, HR, production support	New distribution plan, hiring plan

2. Vision, Mission, Goals & Objectives



Vision

- A vision is a vivid picture of the future the organisation wants to create - inspiring, long-term, short and memorable. Example: "To be the most trusted car maker in India."

Mission

- A mission states the organisation's purpose : what business we are in, whom we serve, how we are different, our values. Example: "We make safe, affordable cars for Indian families through innovation and quality."
- Good mission : clear, realistic, customer-focused, motivating, includes products, markets, technology, philosophy, concern for employees and society

Basis	Vision	Mission
Answers	Where do we want to go?	Why do we exist or what do we do?
Time	Future (10+ years)	Present and near future
Nature	Aspirational	Practical, action-oriented

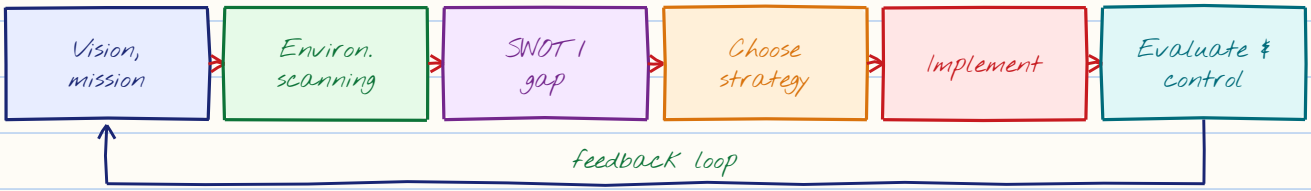
Goals & Objectives

- Goals : broad, qualitative or long-term ends (become market leader) ; Objectives : specific, quantitative and time-bound (raise sales by 15% in FY 2027-28)
- SMART objectives : Specific, Measurable, Achievable, Relevant, Time-bound

- Types : economic (profit, market share), social (CSR), human (training), national (exports) ; Drucker's 8 key result areas : market standing, innovation, productivity, resources, profitability, manager performance, worker performance, public responsibility

3. Corporate Planning Process

Corporate planning = formal, systematic process of deciding the long-range direction of the whole organisation and the means to reach it.



1. Define vision, mission, objectives
2. Environmental scanning - find opportunities and threats ; Internal appraisal - find strengths and weaknesses
3. Strategic gap analysis or SWOT - where we are vs where we want to be
4. Formulate alternative strategies and evaluate them
5. Choose the best strategy - by risk, return, fit, resources
6. Prepare medium and short-range plans, budgets, policies (derived plans)
7. Implement and monitor ; take corrective action (control) - feedback to step 1

4. Environmental Scanning

Environmental scanning = continuously monitoring, collecting and analysing information about forces outside (and inside) the organisation to spot opportunities and threats early (Aguilar).

Environment

- External - Macro (PESTLE) : Political, Economic, Social, Technological, Legal, Environmental
- External - Micro (task or industry) : customers, suppliers, competitors, substitutes, new entrants, banks, govt. regulators - analysed by Porter's 5

Forces

- Internal : resources, people, structure, culture, technology, finance, marketing, R&D → strengths and weaknesses

Techniques

- ETOP (Environmental Threat and Opportunity Profile) - SWOT - PESTLE - Porter's 5 forces - BCG matrix - scenario building - Delphi - industry and competitor analysis

Porter's 5 forces (industry attractiveness)

- Threat of new entrants - Bargaining power of buyers - Bargaining power of suppliers - Threat of substitutes - Rivalry among existing competitors

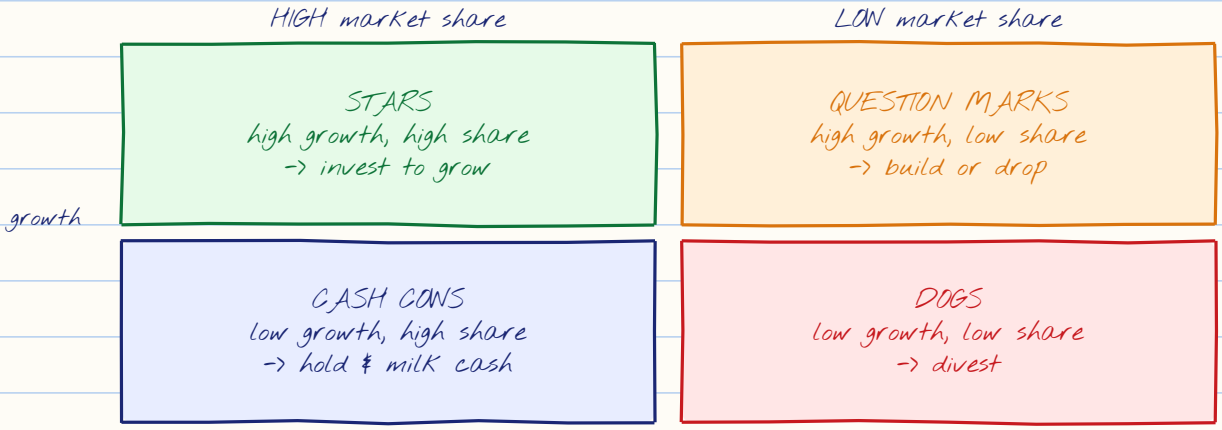
5. Steps in Strategy Formulation & Implementation

Formulation

1. Set or review mission and objectives 2. Environmental and internal analysis (SWOT) 3. Identify strategic alternatives 4. Evaluate alternatives (criteria: fit, feasibility, risk, return) 5. Select strategy 6. Prepare strategic plan

Grand (corporate) strategies - Ansoff / Glueck

- Stability - maintain present business, marginal improvements - Expansion (growth) - market penetration, market development, product development, diversification , mergers and acquisitions, joint ventures
- Retrenchment - turnaround, divestment, liquidation - Combination - mix of the above
- Porter's generic strategies : Cost leadership , Differentiation , Focus
- BCG matrix : Stars (high growth - high share), Cash cows, Question marks, Dogs → invest or hold or harvest or divest



Implementation (put the strategy into action)

- Project implementation (set up facilities, schedules) - Procedural implementation (laws, licences) - Resource allocation (money, people, time)
- Structural implementation (organisation design, delegation) - Behavioural (leadership, culture, motivation, change management) - Functional (policies for marketing, finance, HR, production)
- Control and evaluation : balanced scorecard, KPIs, budgets, audits → corrective action. Framework: McKinsey 7S - Strategy, Structure, Systems, Shared values, Style, Staff, Skills
- Barriers: resistance to change, poor communication, lack of top support, weak resources, short-term thinking

Strategy evaluation - Balanced Scorecard (Kaplan & Norton)

- Financial (profit, ROI) - Customer (satisfaction, retention) - Internal processes (quality, cycle time) - Learning and growth (skills, innovation)
- Evaluate : Are assumptions valid? Is performance on target? Take corrective action (revise strategy, objectives or resources)

6. SWOT Analysis

SWOT (Albert Humphrey) = framework to match internal Strengths and Weaknesses with external Opportunities and Threats and choose strategy.

INTERNAL	EXTERNAL
STRENGTHS (+) brand, skilled staff, technology, cash, location	OPPORTUNITIES (+) growing market, new tech, policy support, export
WEAKNESSES (-) old machines, debt, weak R&D, high attrition	THREATS (-) competition, recession, new laws, tech change

TOWS strategy	With opportunities (O)	With threats (T)
Use strengths (S)	SO : use strengths to grab opportunities (growth)	ST : use strengths to defend threats (diversify)
Reduce weaknesses (W)	WO : overcome weaknesses with opportunities (turnaround)	WT : minimise weaknesses and avoid threats (retrench)

Using SWOT to strengthen a project

- 1. List S, W, O, T with evidence 2. Rank by importance 3. Match (TOWS) 4. Convert into actions with owners, dates 5. Review regularly
- Example (campus food-delivery start-up): S - low-cost student delivery force ; W - small capital ; O - 10,000 hostel students ; T - big apps with discounts → SO : tie-up with mess and canteens, student-only offers ; WT : stay in one campus first

7. Project Management

A project = temporary endeavour with a start and end, a unique output, specific objectives, limited resources and cost (e.g. building a bridge, launching an app). Project management = planning, organising, directing and controlling resources for a project to meet scope, time, cost and quality targets.

Project life cycle

- Conceptualisation or initiation → Planning (WBS, schedule, budget) → Execution → Monitoring and control → Closure or handover
- Tools : Gantt (bar) chart , milestone chart , network (PERT or CPM) , WBS, resource levelling. Triple constraint: scope = time = cost (quality at the centre)

Network analysis - basic terms

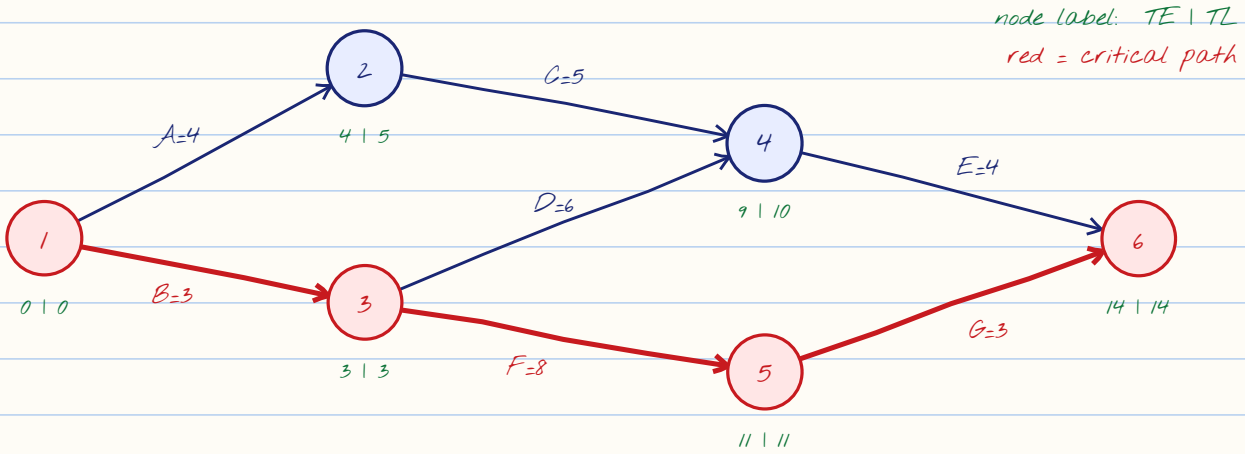
Term	Meaning
Activity	Time- and resource-consuming task, shown by an arrow (→)
Event (node)	Start or finish point of activities, shown by a circle ; consumes no time
Dummy activity	Dotted arrow, zero time - shows logical dependency only
Path	Continuous chain of activities from start to end event
Critical path	Longest path - decides project duration ; zero float
Float or slack	Spare time : Total float = LS - ES = LF - EF ; Free float, Independent float
Predecessor or successor	Activity that must precede or follow another

- Network rules : one start and one end event ; no loops ; events numbered in increasing order ($i < j$) ; no two activities share the same pair of events (use dummy)

Basis	CPM	PERT
Origin	DuPont and Remington Rand, 1957 (construction, chemical)	US Navy, Polaris missile, 1958 (R&D)
Time estimate	Single (deterministic)	Three (a, m, b) - probabilistic
Orientation	Activity-oriented	Event-oriented
Focus	Time-cost trade-off (crashing)	Time control, probability
Use	Repetitive, Known jobs	Non-repetitive, uncertain jobs

8. CPM - Identifying the Critical Path

Steps : draw network → forward pass (ES, EF = ES + t ; event TE = max) → backward pass (LF, LS = LF - t ; event TL = min) → float = LS - ES → critical path = activities with zero float.



Activity	Nodes	t	ES	EF	LS	LF	Float	Critical?
A	1-2	4	0	4	1	5	1	No
B	1-3	3	0	3	0	3	0	Yes
C	2-4	5	4	9	5	10	1	No
D	3-4	6	3	9	4	10	1	No
E	4-6	4	9	13	10	14	1	No
F	3-5	8	3	11	3	11	0	Yes
G	5-6	3	11	14	11	14	0	Yes

Result

- Paths : A-C-E = 13 ; B-D-E = 13 ; B-F-G = 3 + 8 + 3 = 14 days (longest)
  → Critical path = 1-3-5-6 (B-F-G), project duration = 14 days

Types of float (from the example)

Float	Formula	Meaning
Total float	LS - ES (= LF - EF)	Delay possible without delaying the project
Free float	TE(j) - TE(i) - t	Delay possible without affecting the next activity
Independent float	TE(j) - TL(i) - t	Delay possible even if predecessors are late

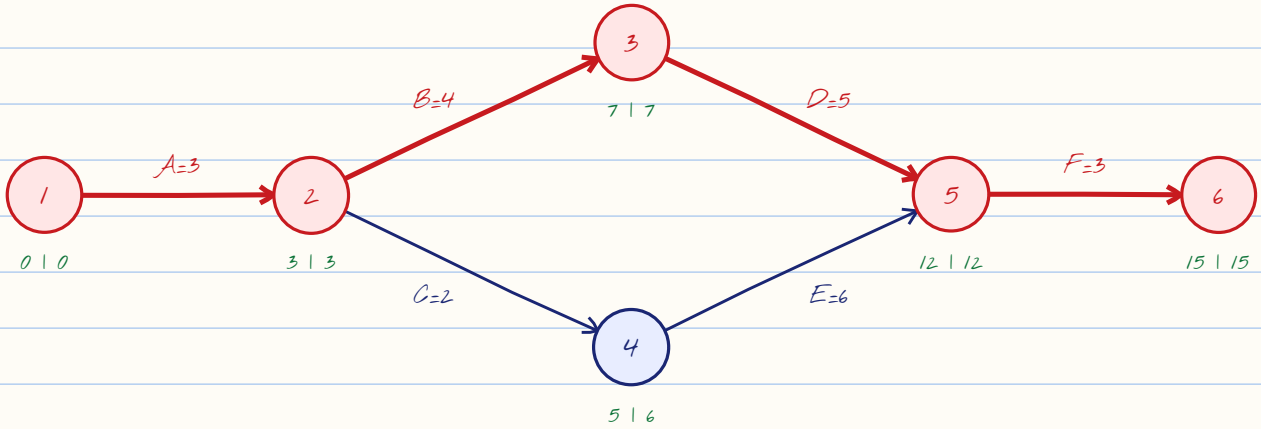
- Example : A → total float 1, free float 4 - 0 - 4 = 0 ; E → total float

1, free float = 14 - 9 - 4 = 1. Critical activities (B, F, G) have all floats = 0.

Second problem - from a precedence table

Activity	Predecessor	Time (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	6
F	D, E	3

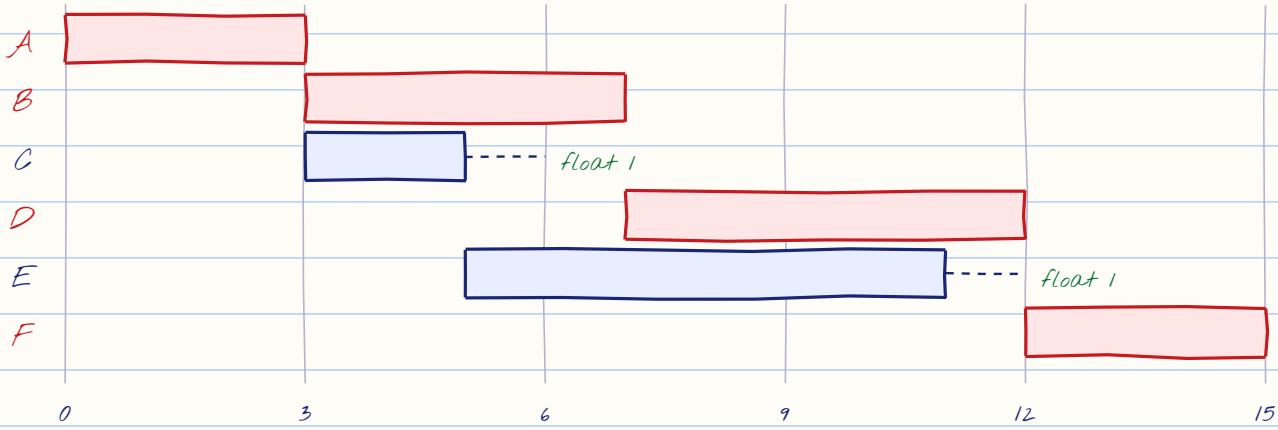
red = critical (A-B-D-F = 15)



Activity	t	ES	EF	LS	LF	Float
A	3	0	3	0	3	0
B	4	3	7	3	7	0
C	2	3	5	4	6	1
D	5	7	12	7	12	0
E	6	5	11	6	12	1
F	3	12	15	12	15	0

- Paths : A-B-D-F = 3 + 4 + 5 + 3 = 15 ; A-C-E-F = 3 + 2 + 6 + 3 = 14 → Critical path A-B-D-F, duration 15 days

Gantt (bar) chart of the same project



- Gantt chart shows when each activity starts and ends but not the dependency clearly ; network shows dependency, critical path and float. PERT or CPM merits : planning, control by exception, resource focus, what-if analysis ; demerits : needs time estimates, complex for large projects.

9. PERT - Probability of Completing in Given Time

Formulae

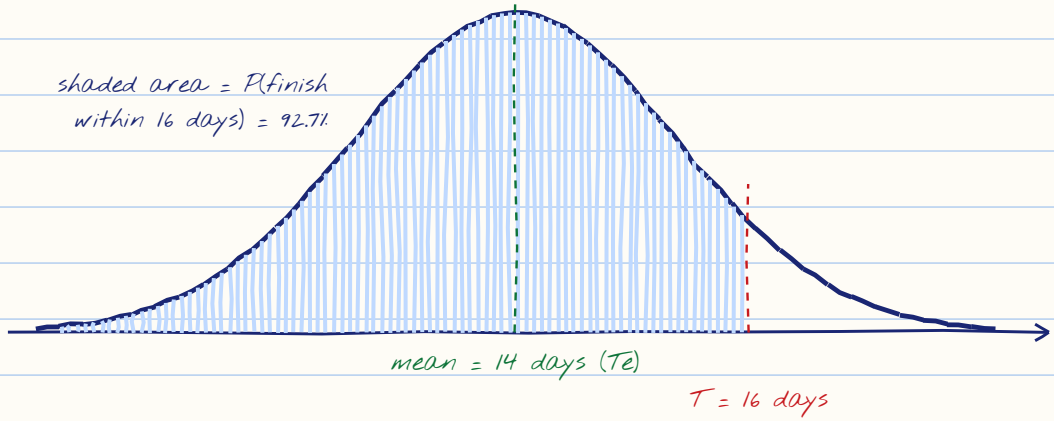
PERT formulae

- Expected time te = (a + 4m + b) ÷ 6 (a = optimistic, m = most likely, b = pessimistic)
- Variance = ((b − a) ÷ 6)² ; Std. deviation = sqrt(variance)
- Project mean Te = sum of te on critical path ; Project variance = sum of variances on critical path
- Z = (T − Te) ÷ sigma → probability from normal table (P = 0.5 when T = Te)

Activity	a	m	b	te	variance
B (1-3)	1	3	5	3	0.444
F (3-5)	5	8	11	8	1.000
G (5-6)	1	3	5	3	0.444
Total (critical)				Te = 14	1.889

Example - same network

- Project variance = 1.889 ; sigma = sqrt(1.889) = 1.374 days
- P(complete within 16 days) : Z = (16 - 14) ÷ 1.374 = 1.455 → P = 0.927 (92.7%)
- P(complete within 12 days) : Z = (12 - 14) ÷ 1.374 = -1.455 → P = 1 - 0.927 = 0.073 (7.3%)
- Duration for 95% confidence = 14 + 1.645 x 1.374 = 16.26 days



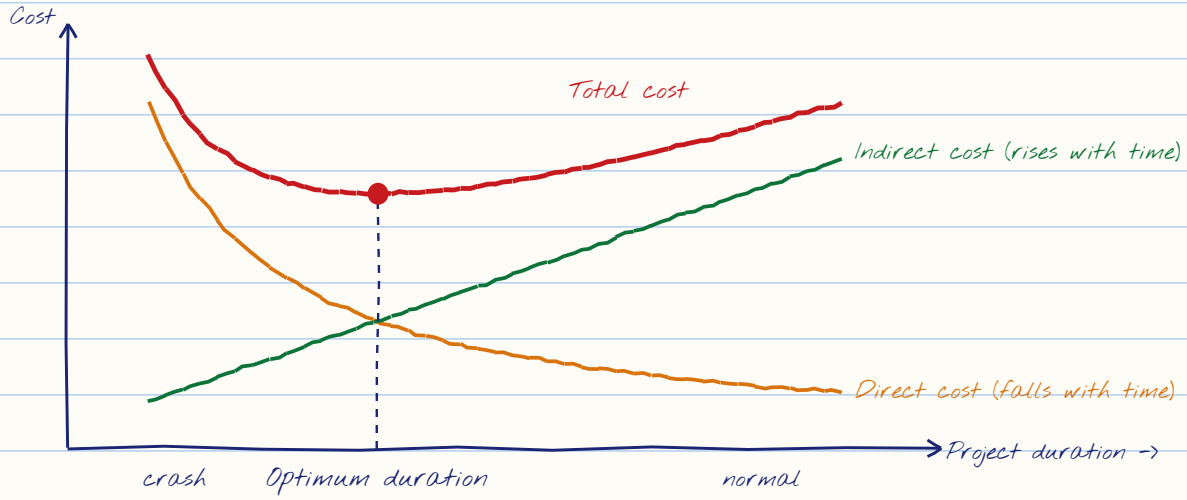
Second PERT problem (critical path A-B-D-F of the 15-day project)

Activity	a	m	b	te	variance
A	2	3	4	3	0.111
B	2	4	6	4	0.444
D	3	5	7	5	0.444
F	2	3	4	3	0.111
Total				15	1.111

- sigma = sqrt(1.111) = 1.054 ; P(finish in 17 days) : Z = (17 - 15) ÷ 1.054 = 1.90 → 0.971 (97.1%) ; P(finish in 14 days) : Z = -0.95 → 0.171 (17.1%)

10. Project Cost Analysis & Crashing

Crashing = shortening project time by adding resources to critical activities at least extra cost. Direct cost rises when time is cut ; indirect cost (overheads, supervision, penalty) falls.



Terms & formula

- Normal time or cost (Tn, Cn) ; Crash time or cost (Tc, Cc)
- Cost slope = (Cc − Cn) ÷ (Tn − Tc) = extra cost per day saved
- Steps : find critical path → crash the critical activity with lowest cost slope (by 1 day each time) → recheck for new critical paths → stop when slope > indirect cost saved per day, or when no activity can be crashed

Activity	Normal (days)	Crash (days)	Normal cost (Rs)	Crash cost (Rs)	Cost slope (Rs/day)
A (1–2)	4	3	400	500	100
B (1–3)	5	3	500	700	100
C (2–4)	5	3	600	900	150
D (3–4)	3	2	300	420	120
E (4–5)	5	4	500	650	150

Network: A(1–2), B(1–3), C(2–4), D(3–4), E(4–5). Paths : A–C–E = 14 days (critical) and B–D–E = 13 days. Direct cost (normal) = Rs.2300. Indirect cost = Rs.130 per day.

Step	Action (crash)	Days	Direct (Rs)	Indirect (Rs)	Total (Rs)
0	Normal schedule	14	2300	1820	4120
1	A by 1 day (slope 100)	13	2400	1690	4090
2	E by 1 day (slope 150, common)	12	2550	1560	4110

Step	Action (crash)	Days	Direct (Rs)	Indirect (Rs)	Total (Rs)
3	B and C by 1 day each (100 + 150)	11	2800	1430	4230

Conclusion

- Total cost is lowest (Rs.4090) at 13 days - beyond this, crashing costs more than the indirect saving (Rs.130/day). Optimum duration = 13 days , optimum cost = Rs.4090 .

11. Important Questions (Unit IV)

1. Define strategy. Explain vision, mission, goals and objectives with examples.
2. Explain the corporate planning process and environmental scanning.
3. Explain the steps in strategy formulation and implementation.
4. What is SWOT analysis? Explain with an example.
5. Differentiate PERT and CPM. Explain network terms.
6. Solve a CPM problem : find ES, EF, LS, LF, float and critical path.
7. PERT problem : find expected time, variance, and probability of completion in given time.
8. Explain project crashing with a simple problem.

Last-minute revision

- te = (a + 4m + b) ÷ 6 ; variance = ((b − a) ÷ 6)² ; Z = (T − Te) ÷ sigma ; float = LS − ES ; critical path = longest path (zero float) ; cost slope = (Cc − Cn) ÷ (Tn − Tc)

UNIT - V

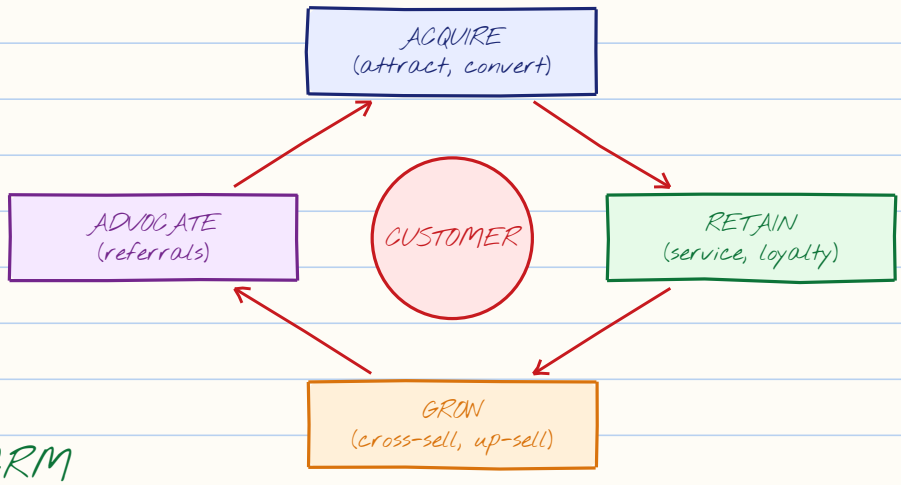
Contemporary Issues in Management

Syllabus at a glance

- CRM - TQM - Six Sigma - SCM - ERP - Performance management - Employee engagement and retention
- BPR and benchmarking - Knowledge management - Change management - Sustainability and CSR

1. Customer Relationship Management (CRM)

CRM = strategy, process and technology to acquire, retain and grow profitable customers by managing every interaction across the customer lifecycle. Goal : customer lifetime value (CLV) instead of one-time sales.



Types of CRM

- Operational : automates sales, marketing and service (contact mgmt, lead tracking, helpdesk) - Analytical : data mining, segmentation, churn prediction, CLV - Collaborative : shares customer data across departments, partners and channels

Components & tools

- Customer database (single view) - Sales force automation - Marketing automation - Call centre or chatbots - Social CRM - Analytics. Tools : Salesforce, HubSpot, Zoho CRM, Microsoft Dynamics

Benefits & challenges

- Benefits : higher loyalty and retention (retaining costs 5-7 times less than acquiring), personalised service, cross-selling, better forecasting, lower cost
- Challenges : high cost, poor data quality, employee resistance, privacy laws (DPDP Act 2023), lack of top-management support

Key idea

- 80% of profit often comes from 20% of customers (Pareto) → identify, reward and retain them. Satisfaction → Loyalty → Advocacy → Profit.

Customer Lifetime Value (CLV) - example

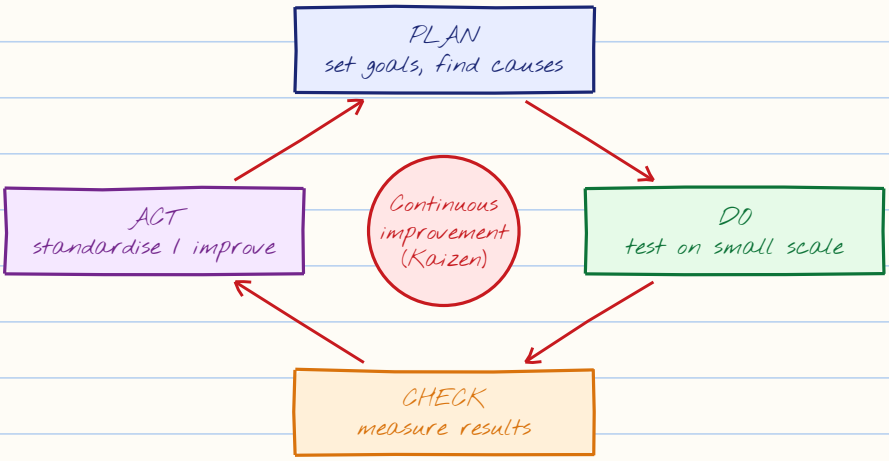
- CLV = average purchase value × purchases per year × years of relationship × profit margin
- Rs.2000 × 6 purchases × 4 years × 20% margin = Rs.9,600 → the firm can afford to spend up to this much to win and keep one customer
- Related terms : customer churn rate , CAC (customer acquisition cost), NPS (Net Promoter Score = % promoters - % detractors)

2. Total Quality Management (TQM)

TQM = organisation-wide approach in which everyone, at every level, is committed to continuously improving quality and meeting customer needs. "Quality is everyone's job."

Principles (core elements)

1. Customer focus (internal and external customers) 2. Top-management leadership and commitment 3. Total employee involvement (quality circles, empowerment)
4. Process approach 5. Continuous improvement (Kaizen) 6. Fact-based decisions (SPC, data) 7. Supplier partnership 8. Prevention rather than inspection (get it right first time)



Quality gurus

Guru	Contribution
W. Edwards Deming	14 points ; PDCA cycle ; 85% of problems are due to the system (management), not workers
Joseph Juran	"Fitness for use" ; Quality Trilogy : planning, control, improvement
Philip Crosby	"Quality is free" ; Zero defects ; do it right the first time
Ishikawa	Fishbone diagram ; quality circles ; Taguchi - loss function ; Feigenbaum - total quality control

Deming's 14 points (condensed)

- Create constancy of purpose ; adopt the new philosophy ; stop mass inspection ; do not award business on price alone ; improve constantly ; train on the job ; institute leadership
- Drive out fear ; break down barriers between departments ; remove slogans and numerical quotas ; remove barriers to pride of workmanship ; encourage education and self-improvement ; put everybody to work on the transformation

5S	Japanese	Meaning
Sort	Seiri	Remove unnecessary items
Set in order	Seiton	A place for everything
Shine	Seiso	Clean the workplace
Standardise	Seiketsu	Make rules and checklists
Sustain	Shitsuke	Discipline, Keep the habit

- Quality circle : small voluntary group (6-10) from the same work area

meeting regularly to solve quality problems (Japan, Ishikawa). ISO 9001 : international QMS standard (certification by audit, valid 3 years). Cost of quality : prevention + appraisal + failure (internal and external).

Tools & benefits

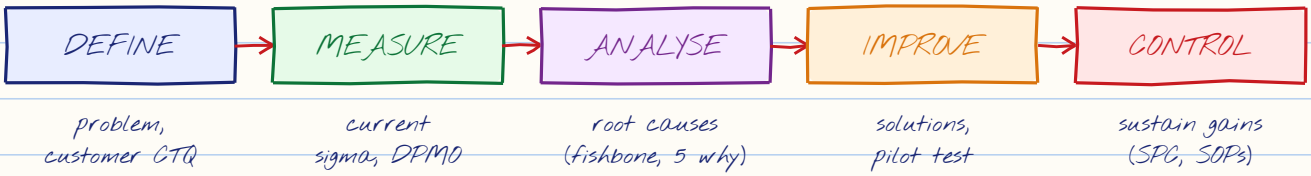
- 7 QC tools : check sheet, histogram, Pareto chart, cause-and-effect (fishbone), scatter diagram, control chart, flow chart ; also 5S, Kaizen, Poka-yoke, QFD, benchmarking, ISO 9001
- Benefits : fewer defects and rework, lower cost, customer satisfaction, higher morale, competitive edge - Barriers : lack of commitment, resistance, no training, short-term view, poor measurement

3. Six Sigma Concept

Six Sigma = data-driven, disciplined methodology (Motorola, Bill Smith, 1986; popularised by GE - Jack Welch) to reduce variation and eliminate defects so that a process produces only 3.4 defects per million opportunities (DPMO). Sigma = standard deviation.



Sigma level	Yield (%)	DPMO	Meaning
1 sigma	30.9	691,462	very poor
2 sigma	69.1	308,538	poor
3 sigma	93.3	66,807	average industry
4 sigma	99.38	6,210	good
5 sigma	99.977	233	very good
6 sigma	99.99966	3.4	world class



- DMAIC (existing process) : Define, Measure, Analyse, Improve, Control ; DMADV or DFSS (new process) : Define, Measure, Analyse, Design, Verify
- Belts : White, Yellow, Green (part-time project leaders), Black (full-time experts), Master Black Belt (coach), Champion (executive sponsor)
- Lean Six Sigma = Lean (remove waste : defects, overproduction, waiting, transport, over-processing, inventory, motion, unused talent) + Six Sigma (remove variation)

Example - DPMO

- 500 units checked, 5 opportunities for defect per unit, 25 defects found
- DPMO = defects ÷ (units x opportunities) x 1,000,000 = 25 ÷ (500 x 5) x 10^6 = 10,000 → about 3.8 sigma

Root-cause tools used in Analyse phase

- 5 Whys : machine stopped → fuse blown → bearing overloaded → no lubrication → pump worn out → no maintenance schedule (root cause)
- Fishbone (Ishikawa) : causes grouped under Man, Machine, Material, Method, Measurement, Environment (6 M's) leading to the effect (defect)

4. Supply Chain Management (SCM)

Supply chain = network of suppliers, manufacturers, warehouses, distributors, retailers and customers that converts raw materials into finished goods delivered to the end customer. SCM = planning, coordinating and controlling the whole chain - flows of materials, information and finance - to deliver maximum customer value at lowest total cost.

MATERIAL / PRODUCT flow ->



<- INFORMATION and CASH flow

Plan - Source - Make - Deliver - Return (SCOR model)

Objectives & Key decisions

- Right product, right place, right time, lowest cost - Reduce inventory and lead time - Improve service level and flexibility - Build supplier partnership - Visibility and responsiveness
- Drivers (Chopra) : facilities, inventory, transportation, information, sourcing, pricing

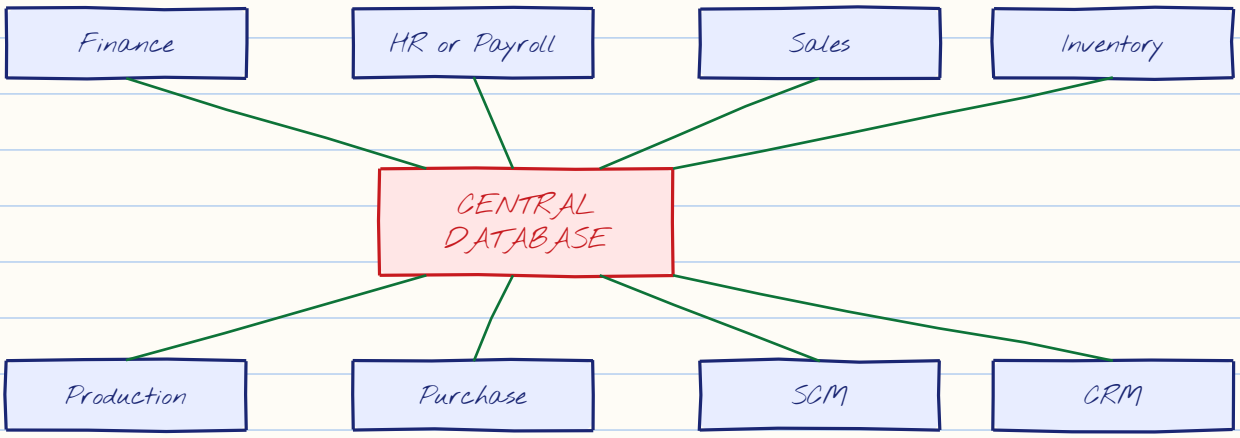
Important concepts

- Bullwhip effect : small change in customer demand creates larger swings upstream due to poor information sharing → solved by information sharing, VMI, POS data
- JIT or lean - VMI (vendor managed inventory) - Cross-docking - 3PL or 4PL logistics - Make-to-stock vs make-to-order - e-SCM, RFID, blockchain traceability - Reverse logistics
- Benefits : lower cost, shorter cycle time, less stock, better quality and customer service. Challenges : lack of trust, disruption risks (COVID, wars), complexity, IT integration. Example : Amazon, Walmart, Toyota, Reliance Retail

Basis	Push strategy	Pull strategy
Trigger	Forecast of demand	Actual customer orders
Inventory	High (make-to-stock)	Low (make-to-order, JIT)
Risk	Over-stocking, obsolescence	Stock-outs, longer lead time
Example	FMCG, soaps, biscuits	Dell computers, Toyota JIT

5. Enterprise Resource Planning (ERP)

ERP = integrated software suite that connects all functions (finance, HR, production, sales, purchase, inventory) on a single database so that data is entered once and shared in real time.



Evolution

- MRP (1960s, materials) → MRP-II (1980s, + capacity, finance) → ERP (1990s, whole enterprise) → ERP-II or cloud ERP and AI (2000s +)

Features, benefits, limits

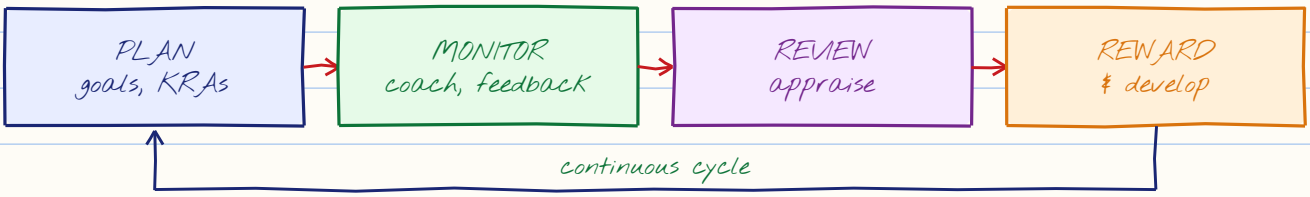
- Integrated, real-time, single database, process-oriented, modular, best-practice templates, multi-currency, role-based access
- Benefits : no duplication, faster decisions, lower inventory, better control and transparency, improved customer service
- Limitations : very costly, complex, long implementation (6-24 months), employee resistance, customisation difficulty, vendor lock-in, data security

Module	What it handles
Finance and accounting	General ledger, accounts payable or receivable, budgeting, GST and tax, costing
Human resources	Payroll, attendance, recruitment, training, appraisal, employee self-service
Production (MRP)	BOM, production planning, scheduling, shop-floor control, quality
Materials and purchase	Vendor management, purchase orders, inventory, warehouse
Sales and distribution	Quotation, orders, billing, dispatch, CRM link

Aspect	Details
Vendors	SAP S/4HANA, Oracle NetSuite or Fusion, Microsoft Dynamics 365, Infor, Odoo (open source), Tally Prime (SMEs)
Implementation phases	Planning and vendor selection → Business process mapping (as-is, to-be) → Customisation → Data migration → Testing → Training → Go-live → Support
Go-live strategies	Big bang (all at once) ; Phased (module by module) ; Parallel (old + new together) ; Pilot

6. Performance Management

Performance Management (PMS) = continuous process of setting goals, monitoring, coaching, reviewing and rewarding performance to align individual work with organisational strategy. (PA = once a year review ; PMS = whole year cycle, forward looking.)



- KRA (Key result areas) = what the job must achieve ; KPI = measurable indicators (sales growth, defect rate, on-time delivery) ; OKR (Objectives and Key Results) ; Balanced Scorecard (financial, customer, internal process, learning)
- Good PMS : clear goals, regular feedback, 360-degree input, fair rating, links to pay, promotion, training and career development

7. Employee Engagement & Retention

Engagement = employee's emotional commitment to the organisation and its goals - willing to give discretionary effort . (Gallup : engaged or not engaged or actively disengaged.)

Basis	Satisfaction	Engagement
Meaning	Contentment with job, pay, conditions	Passion and commitment to give extra effort
Result	Stays, does the required work	Performs beyond expectations, innovates

Drivers of engagement

- Meaningful work and autonomy - Good leadership and trust - Recognition and rewards - Career growth and learning - Work-life balance - Open communication - Fair pay - Team culture and wellbeing

Retention

- Retention = efforts to keep valued employees and reduce turnover. Turnover rate (%) = (number of separations ÷ average number of employees) × 100. Example : 30 leave, average staff 300 → 10%
- Strategies : competitive pay and benefits (ESOP), career paths, training, flexible or hybrid work, recognition, good onboarding, stay interviews, exit interviews, wellness programmes, employer branding
- Cost of turnover : hiring, training, lost productivity and knowledge, low morale (often 50-200% of annual salary)

8. BPR & Benchmarking

Business Process Re-engineering (BPR)

Hammer and Champy (1993) : "the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in cost, quality, service and speed." (Not small, gradual - like Kaizen.)

- Principles : organise around outcomes, not tasks ; let the user of the output do the work ; capture information once at the source ; treat dispersed resources as centralised ; link parallel activities ; use IT as enabler
- Steps : identify processes → map as-is → find problems → design to-be radical process → implement (pilot) → monitor
- Examples : Ford accounts payable (500 → 125 clerks), IBM Credit, Taco Bell. Risks : job loss, resistance, high failure rate (~50-70%), poor leadership

Basis	BPR	Kaizen or TQM
Change	Radical, breakthrough	Small, continuous
Scope	Cross-functional process	Everyone, every day
Risk	High	Low

Benchmarking

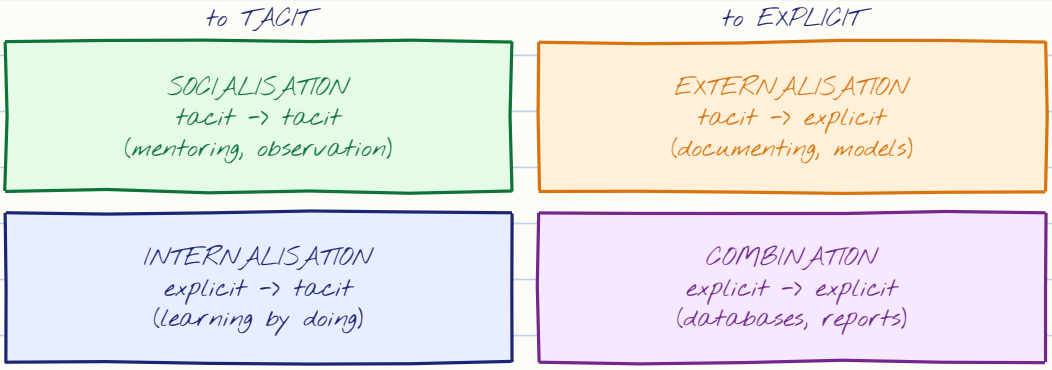
Benchmarking = continuous process of measuring products, services and practices against the best competitors or industry leaders to learn and improve. (Xerox, 1979 - Robert Camp.)

- Types : Internal (between own units), Competitive , Functional (same function, different industry), Generic (best process anywhere), Strategic or performance benchmarking
- Process : Plan (what to benchmark) → Find partner → Collect data → Analyse gap → Adopt and implement → Monitor . Benefits : set realistic targets, avoid reinventing the wheel, spot best practices

9. Knowledge Management (KM)

KM = systematic process of creating, capturing, storing, sharing and applying knowledge to improve organisational performance and innovation. Data → Information → Knowledge → Wisdom (DIKW).

Type	Tacit Knowledge	Explicit Knowledge
Nature	In people's heads, experience, intuition	Written, codified, documented
Sharing	Hard - by mentoring, conversation	Easy - manuals, databases
Example	Skill of a master craftsman	SOP, patent, report



- Nonaka and Takeuchi's SECI model : the spiral of knowledge conversion shown above
- KM process : Create → Capture → Organise and store → Share → Apply → Review . Tools : intranet, wikis, databases, expert directories, communities of practice (CoP), lessons-learned, AI search
- Benefits : faster learning, avoid repeat mistakes, innovation, preserves know-how when people leave. Barriers : Knowledge hoarding, no trust or reward, technology gap

10. Change Management

Change management = structured approach to moving individuals, teams and organisations from the current state to a desired future state while minimising resistance.

Forces & resistance

- Forces for change : technology, competition, globalisation, customer needs, laws, crisis, new leadership
- Why people resist : fear of the unknown and job loss, habit, loss of power or status, poor communication, lack of trust, "what's in it for me?"

Models



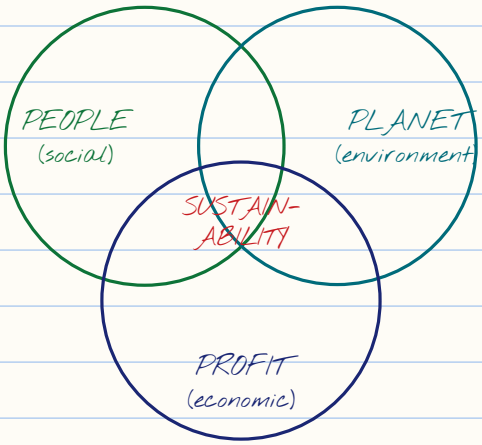
Model	Steps
Lewin (1951)	Unfreeze (create need, force-field analysis) → Change (move to new state) → Refreeze (reinforce, new norms)
Kotter's 8 steps	Create urgency → Build guiding coalition → Form vision → Communicate → Remove barriers → Create short-term wins → Build on gains → Anchor in culture
ADKAR (Prosci)	Awareness, Desire, Knowledge, Ability, Reinforcement

Overcoming resistance (Kotter & Schlesinger)

- Education and communication - Participation and involvement - Facilitation and support - Negotiation and agreement - Manipulation or co-optation - Explicit or implicit coercion (last resort)

II. Sustainability & Corporate Social Responsibility

Sustainable development (Brundtland, 1987) = meeting the needs of the present without compromising the ability of future generations to meet their own needs.



Triple Bottom Line
(John Elkington, 1994)

- ESG (Environmental, Social, Governance) investing and reporting ; SEBI's BRSR (Business Responsibility and Sustainability Report) mandatory for top 1000 listed companies in India
- UN SDGs : 17 goals to be achieved by 2030 (no poverty, quality education, clean energy, climate action...) - Circular economy (reduce, reuse, recycle) - Green management : renewable energy, waste reduction, carbon footprint, ISO 14001, net-zero targets
- CSR (Companies Act 2013, Section 135) : companies with net worth >= Rs.500 cr or turnover >= Rs.1000 cr or net profit >= Rs.5 cr spend 2% of average net profit (last 3 years) on CSR ; CSR committee ; activities in Schedule VII (education, health, hunger, environment, rural development)
- Benefits of sustainability : brand value, cost saving (energy), investor confidence, risk reduction, talent attraction, legal compliance. Examples : Tata (Tata Trusts, TCS), Infosys (carbon neutral), ITC (water positive), Mahindra

12. Important Questions (Unit V)

1. What is CRM? Explain its types, benefits and components.
2. Explain TQM and its principles. Discuss Deming's contribution.
3. What is Six Sigma? Explain DMAIC with an example.
4. Explain supply chain management and the bullwhip effect.
5. What is ERP? Explain its modules, benefits and implementation.
6. Differentiate performance appraisal and performance management.
7. Explain employee engagement and retention strategies.
8. Write notes on BPR, benchmarking, Knowledge management.
9. Explain change management models. How do you overcome resistance?
10. Discuss sustainability and CSR with the triple bottom line.

Last-minute revision

- CRM = acquire, retain, grow ; TQM = PDCA + Kaizen ; Six Sigma = 3.4 DPMO, DMAIC ; SCM = bullwhip ; ERP = single database ; BPR = radical ; Lewin = unfreeze-change-refreeze ; TBL = people, planet, profit